**Institute of**
**Digital and Autonomous Construction**
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IDAC TUHH

RESEARCH PROJECT

for the degree program of M.Sc. Information and Communication Systems

# Localization of mobile robots coupling building information modeling and Wi-Fi-based positioning

*Lokalisierung mobiler Roboter durch Kopplung von Building Information Modeling und Wi-Fi-basierter Positionierung*

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Matriculation number: 565772
January 1, 2025

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## Abstract

Mobile robots are increasingly being utilized in indoor spaces for automated inspections and monitoring tasks. Reliable and accurate indoor localization is therefore essential. However, conventional indoor localization techniques, such as radio frequency identification (RFID), Bluetooth, or wireless fidelity (Wi-Fi), received signal strength indicator (RSSI), are prone to limited robustness and accuracy due to signal attenuation and multipath propagation. Existing techniques neglect key built environmental factors, including materials, layout and structure, which widely affect access point (AP) placement and signal propagation. Therefore, achieving centimeter-level accuracy in indoor positioning with Wi-Fi signals continues to be a significant hurdle. Additionally, lack of optimization strategies for determining the deployment and optimal number of AP hinders reliable positioning accuracy and signal coverage. To address such challenges, the work proposes a Building information modeling (BIM)-assisted indoor localization framework that integrates IEEE 802.11az physical layer characteristics with fusion and optimized access point placement. The research project introduces a MATLAB software framework that utilizes a hybrid technique, fusing multiple physical-layer characteristics, derived from IEEE 802.11az Wi-Fi signals, into an integrated input for a convolutional neural network (CNN) for precise learning-based robot localization. Simultaneously, a custom genetic algorithm (GA) is developed to enhance the automatic placement of APs through a discrete optimization strategy in a given infrastructure model, to increase localization coverage. The entire framework, such as feature extraction, fusion of physical-layer characteristics, data generation and CNN-based positioning, is implemented as a simulation on a standard platform. The system does not work in real time; however, operates in batches, enabling various environments to be tested using standard tessellation language (STL) as input. The proposed framework is validated utilizing a BIM-based indoor environments. Performance is estimated under bandwidth impact assessment, effect of station separation, antenna scaling, and material sensitivity. Regression head is evaluated based on RMSE, median and percentile positioning errors summarized from CDF, while classification head is assessed utilizing confusion matrices. Validation outcomes recognize the 2 x 1 antenna configuration, 0.5 m separation distance, and CBW160 MHz bandwidth as the best optimal configuration for compensating hardware complexity and positioning reliability. The proposed multimodal framework substantially enhances robustness over baseline models, obtaining centimeter-level accuracy continues to be a challenge because of the complex multipath propagation and signal noise floor is in-built in dense indoor environments. Eventually the proposed framework achieves a best-case of RMSE of 4.43 m, offering a solid BIM foundation for autonomous robot navigation in indoor environment.

## Eidesstattliche Erklärung

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## Statutory Declaration

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| Degree program: | M.Sc. Information and Communication Systems |
| Matriculation number: | 565772 |

## Topic

Localization of mobile robots coupling building information modeling and Wi-Fi-based positioning
*Lokalisierung mobiler Roboter durch Kopplung von Building Information Modeling und Wi-Fi-basierter Positionierung*

## Introduction

Mobile robots are increasingly deployed for automated inspection of the built environment. To effectively inspect the built environment, reliable centimeter-level localization of mobile robots using building information modeling (BIM) is crucial. Accurate localization of mobile robots enables consistent navigation, repeatable inspections and measurements, and precise location-referencing of inspection findings. The IEEE 802.11az standard, also known as Next-Generation Positioning, introduces physical-layer features that enhance wireless fidelity (Wi-Fi) ranging and positioning accuracy. The improvements in ranging and positioning accuracy may be utilized for time-of-arrival and angle-of-arrival estimation to provide high resolution and accuracy in robot localization. In this student project, a framework for 3D indoor localization coupling BIM models and Wi-Fi communication standards must be developed. State-of-the-art techniques for 3D indoor positioning using Wi-Fi-based communication are to be reviewed. The framework must be implemented and validated in simulated environments with varying geometries across numerous BIM models.

## Tasks

1. Perform a literature review of the state-of-the-art techniques for 3D indoor positioning using W i-Fi communication with an emphasis on IEEE 802.11az. Analyze state-of-the-art methods, identify challenges, and gaps in the current technology related to BIM-based robot localization.

2. Based on the literature review in Task 1, develop a framework that is capable of 3D indoor positioning using the 802.11az standard for WiFi communication. The framework must be capable of fusing position estimates from robot odometry and WiFi-based positioning.
3. Implement the 3D indoor positioning framework by developing the requisite software modules.
4. Validate the framework implemented in Task 3 by testing indoor positioning of a mobile robot in simulation environments across BIM models.
5. Discuss the results of the validation tests and the localization accuracy of the framework. Discuss accuracy limits, failure cases, and opportunities for future improvement. Provide an outlook on the applicability of the framework in practice.

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|---|---|
| **Start:** | 01.09.2025 |
| **Submission:** | 01.03.2026 |
| **First examiner:** | Professor Dr. Kay Smarsly |
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# Acronyms

$B_{\textbf{box}}$ Bounding Box. 29, 30

$P_{\textbf{grid}}$ grid point. 30

**2-D** two-dimensional. 10, 15, 30, 32

**3-D** three-dimensional. 3, 4, 11, 13, 16, 20–22, 27, 28, 31, 36, 38, 40, 48, VIII

**AoA** angle of arrival. 1–4, 7–11, 29–33, 36, 39, 44, 46–48, IX

**AP** access point. 1–5, 8, 10, 12–23, 27–31, 36–38, 40, 41, 47, 48, I, V, VIII, IX

**AWGN** additive white Gaussian noise. 13, 23

**BIM** Building information modeling. 3–5, 8, 11, 15, 19, 21, 22, 30, 38, 40, 45, 47, I, V, VIII

**BLE** Bluetooth low energy. 6

**CDF** cumulative distribution function. 39, I

**CIR** channel impulse response. 1–4, 7–10, 13, 21–25, 27–35, 39, 41, 45–48, V, VIII, IX

**CNN** convolutional neural network. 2–4, 10, 13, 14, 24, 25, I, V, VI, VIII

**CP** cyclic prefix. 33

**CSI** channel state information. 2, 3, 7, 8, 10, 32

**FTM** fine timing measurement. 2, 7, 8

**GA** genetic algorithm. 3, 4, 8, 12–15, 19, 20, 22, 27, I, V, VIII

**GDOP** geometric dilution of precision. 3, 10, 12, 16, 17, 44, V

**GHz** gigaHertz. 2, 7, 22, 27

**GNSS** global navigation satellite system. 1

**HE** high efficiency. 13

**HE-LTF** high efficiency long training fields. 13, 23

**IDAC** Institute of Digital and Autonomous Construction. 20, 22, 36, VIII

**IP** Indoor positioning. 1

**K-nn** k-nearest neighbor. 10

**LiDAR** light detection and ranging. 1, 6

**LoS** line-of-sight. 1, 3, 6, 11, 21, 24, 30, 40, VIII

**LTF** long training field. 7, 8

**MIMO** multiple input multiple output. 6, 20, 22

**MLP** multilayer perceptron. 10, VIII

**MOGA** multi-objective genetic algorithm. 3–5, 15–18, 38, 44, 48, V, VIII

**MSE** mean squared error. 36

**MTL** multi-task learning. 35

**MU-MIMO** multiple users-multiple inputs multiple outputs. 6

**NGP** next generation positioning. 2, 6, 7

**NLOS** non-line-of-sight. 1, 7–10, 12, 40

**NSGA-II** non-dominated sorting genetic algorithm II. 18

**OFDM** orthogonal frequency division multiplexing. 33

**OFDMA** orthogonal frequency division multiple access. 6

**PDOP** position dilution of precision. 19

**ResNet** residual network. 24, 25, 30, 33–35, 37–39, 41, 42, 48, VI, VIII

**RF** radio frequency. 10

**RFID** radio frequency identification. 6, I

**RMSE** root mean square error. 39, 40, 42, 44–47, I

**RSSI** received signal strength indicator. 1, 2, 4, 7, 8, 10, 16, 29–33, 39, 40, 46–48, I, IX

**RTT** round trip time. 7, 8

**SISO** single-input single-output. 44

**SLAM** simultaneous localization and mapping. 1, 6

**SNR** signal-to-noise ratio. 3, 8, 9, 12, 21, 23, 29, 31, 41

**STA** station. 13–17, 20, 21, 23, 27–31, 34, 38, 40, 41, 43, 44, 46, 48, V, VI, VIII, IX

**STL** standard tessellation language. 20, 22, 27–30, 38, 40, 46, I

**SVM** support vector machines. 10

**ToF** time-of-flight. 1–4, 7–11, 29–33, 36, 39, 42, 46–48, IX

**TUHH** Technical University of Hamburg. 20, 22, VIII

**UWB** Ultra wideband. 6

**Wi-Fi** wireless fidelity. 2, 3, 6–8, 10, 22, I, V, IX

**WLAN** wireless local area network. 12, 22, 27

# 1 Introduction

Autonomous mobile robots significantly enhance operational efficiency in automated indoor inspections [4]. Automated inspections require reliable and continuous positioning to enable robotic navigation in cluttered building environments and to perform tasks with minimal human intervention. Unlike outdoor environments, where global navigation satellite system (GNSS) offer an efficient solution, indoor spaces impose restrictions on the achievable accuracy due to signal degradation, multipath propagation, and non-line-of-sight (NLOS) conditions, making precise localization much more challenging. To overcome such challenges, scholars have examined different signal-based techniques. However, major challenges remain in current state-of-the-art systems.

## 1.1 Motivation and background

State-of-the-art localization techniques rely on individual signal features, such as received signal strength indicator (RSSI), time-of-flight (ToF), angle of arrival (AoA), or channel impulse response (CIR). When used in isolation, signal characteristics may be susceptible to environmental dynamics, further degrading localization robustness. To illustrate, RSSI may vary due to human gestures or interference; meanwhile, ToF and AoA require high accuracy and are susceptible to multipath fading [25]. In CIRs, material properties of environments must be known before system deployment[20]. Another aspect of infrastructure design that remains unaddressed is determining the optimal deployment and number of access point (AP)s required to guarantee coverage and precise localization within buildings. Such design aspects that may have been overlooked during building design may reduce positioning reliability and prevent mobile robots from achieving centimeter-level accuracy in indoor environments.

Indoor positioning (IP) systems research utilizes two main techniques, such as onboard sensors and wireless signals from buildings, each with distinct advantages and disadvantages. Onboard techniques using inertial measurement units and odometry provide quick location updates; however increase unbounded drift without external correction mechanisms. Vision-based and light detection and ranging (LiDAR)-based simultaneous localization and mapping (SLAM) systems handle drift accumulation by creating environmental maps and identifying revisited locations, achieving maximum efficiency when environmental conditions provide clear line-of-sight (LoS), stable lighting, and appropriate scene texture. However, performance of vision-based and LiDAR systems suffers in cluttered environments that include variable lighting, moving obstacles, featureless corridors, transparent partitions, and reflective surfaces, while LiDAR systems incur significant computational overhead and hardware costs [2]. Onboard sensor data, combined with wireless signals, enhances robustness despite varying lighting conditions and provides reliable localization throughout the building. Evaluating the physical limitations of the hardware and the behavior of the radio channel is important for assessing the localization frameworks explored in the subsequent section.

## 1.2 Related work

Traditional wireless fidelity (Wi-Fi) positioning systems utilize RSSI, which is important for constructing a map of signal patterns across locations [25]. Since most smartphones can accurately measure signal strength and run adequately, smartphones are usually used for room- or zone-level classification. However, signal strength parameters are irregular and may vary with environmental conditions, such as furniture rearrangement, weather, or human interference, making localization systems unpredictable over time and hard to generalize across different buildings [25]. To achieve higher accuracy than techniques based solely on signal strength, time-based and angle-based systems have been developed. ToF computes transmission time, while AoA estimates the direction of the arrival of signals using multiple antennas. ToF and AoA-based techniques deliver stronger geometric configurations and achieve greater accuracy, particularly with broader signal bandwidths and tuned antenna arrays. Although ToF and AoA techniques have advantages, both techniques suffer when signals reflect walls rather than communicating directly, and generate false angle readings. Therefore, strategic placement of multiple APs and smart selection algorithms ensures system robustness, in addition to mitigating signal reflection [24], [8]. Wi-Fi frameworks have gradually enhanced physical-layer feature capabilities, allowing stable ToF and AoA estimation. The IEEE 802.11 mc standard [1] establishedfine timing measurement (FTM) for accurate distance ranging. In recent times, IEEE 802.11 az, also known as next generation positioning (NGP), has introduced numerous advancements, including secure training signals, coordinated measurements across different APs, and operation in the 6 gigaHertz (GHz) frequency band with wider bandwidth. The advancements offer precise time measurements, improve angle detection, and enhance channel information for machine learning models [17], [21].

Another research technique uses comprehensive CIR or channel state information (CSI) as high-dimensional inputs, analogous to image input in deep learning models. Rather than extracting single quantities, such as distance or signal strength, the proposed technique records the complete signal patterns across frequency components, antenna elements, and time delays. convolutional neural network (CNN) can explore such patterns to learn multipath models, recognizing how signals reflect under varying conditions. The fusion of signal characteristics (RSSI, ToF, AoA, and CIR technique surpasses the technique that employs an individual measurement type or signal characteristics, provided the indoor localization system is accurately calibrated, and noise is eliminated [25]. Signal reflections rely mostly on building infrastructure and materials; therefore, developing comprehensive training materials is crucial. Building information modeling (BIM) fused with ray tracing has emerged as an effective solution [14]. The ray tracing technique employs three-dimensional (3-D) building models to simulate how radio signals travel multiple reflections, direct routes, single bounces, and propagation through materials. The simulation calculates realistic power levels and signal delays based on the communication of radio waves with the environment. BIM-based deterministic ray tracing generates signal patterns with improved accuracy for realistic building environments compared to frameworks depending on isolated signal characteristics, regardless of material adaptation and diffuse scattering complexities [9].

The infrastructure layout affects positioning accuracy due to the spatial design of narrow corridors, furniture, and walls, which generate distinct multipath scenarios that considerably modify radio signal transmission [14]. Conventional Wi-Fi frameworks have been developed to increase data speed and coverage; however, positioning systems require specialized AP configurations, particularly in geometries that minimize position volatility and deliver diverse angular readings. As the evaluation of each possible set of combinations of AP locations is infeasible, discrete optimization techniques [12], especially genetic algorithm (GA) and complementary optimization techniques are employed [5]. A multi-objective genetic algorithm (MOGA) serves the environment as a discrete search space, simultaneously enhances coverage quality, geometric design, and infrastructure cost by merging all purposes into an individual fitness score. By integrating a penalty for using too many APs, the algorithm automatically detects where to deploy APs and how many to deploy, maximizing positioning accuracy rather than coverage [5]. Reliable indoor localization requires a comprehensive technique to overcome signal degradation caused by multipath interference and complex building design. State-of-the-art indoor positioning systems incorporate four key properties, which are **BIM** for environmental modeling, **realistic ray tracing** for signal propagation representation, **GA**-based frameworks optimization, and multi-modal deep learning for position assessment. BIM geometric configurations are fed as input to ray tracing engines, which calculate LoS paths, physical reflection, and diffractions to create site-specific CIRs. ToF and AoA features are extracted from synthetic multi-path environments. GA analyzes the discrete space of user AP locations, assessing fitness functions that balance comprehensive coverage, deployment cost, and geometry metric utilizing geometric dilution of precision (GDOP), which is referred to as the knee point[10]. The associated enhanced deployments feed training information to multi-modal fusion neural models. CNNs derive spatial patterns from CSI tensors, while multi-layer perceptrons encode the received signal indicator, ToF, and AoA readings. Dual-head learning models generate both regression and classification results. The regression head provides accurate position coordinates, while the classification head delivers robust localization. Mutual training between the regression and classification heads improves generalization across various building designs and fluctuating signal-to-noise ratio (SNR) conditions, ultimately establishing a navigation framework capable of achieving high accuracy in enclosed or cluttered office areas [23]. Even though signal standards have evolved considerably, an integrated technique that fuses optimized AP deployment with high-dimensional signal fusion remains missing. The current project tackles the identified research gap by incorporating BIM-based deterministic ray tracing with MOGA framework optimization and multi-modal deep learning. Integrating physical building information with optimized AP deployment ensures that the framework produces a robust training dataset that reflects actual behavior. Multi-modal fusion of RSSI, ToF, AoA, and CIR characteristics into a dual-head CNN enables simultaneous coordinate regression and room/zone classification, thereby achieving stable positioning in cluttered indoor environments.

## 1.3 Objectives and approach

Building on research in multimodal sensing, architecture optimization, and deep learning, this project proposes a unified framework that addresses gaps in the state-of-the-art indoor positioning models. The main methodology comprises high-dimensional signal characteristics from 802.11 az transmissions to build a detailed illustration of the indoor environment. The framework combines multiple signal characteristics into a single integrated input for robot positioning. Although combining multiple physical-layer characteristics delivers the important detailed information, the physical stability of the framework is further ensured by applying GA-based AP deployment optimization.

Systematic deployment of APs confirms that the final training dataset captures actual multi-path behavior and material impact within the 3-D building model. A tailored CNN is designed to perform both regression and classification tasks, enabling the network to estimate continuous X and Y position coordinates while also classifying discrete zones. The regression head enhances positioning accuracy by estimating continuous positions, while the classification head is trained to predict the target device's reference locations or discrete zones within the mapped environment. By training dual-head deep learning models, the proposed framework achieves robustness and strong generalization across different environments and materials.

To optimize the deployment of the proposed framework, a discrete optimization strategy exploiting MOGA is used to estimate the optimal number of APs and the associated locations over a defined user grid. The MOGA aims to balance network coverage, deployment cost, and geometric dilution to produce AP layouts particularly tailored for positioning performance. For each GA user configuration, ray-tracing is performed to synthesize associated signal characteristics and CIRs, verifying that the CNN is trained on information that reflects both the discrete AP deployment and the representing radio-signal propagation. The primary objective of this project is to achieve sub-meter positioning accuracy, aiming for centimeter-level localization in indoor environments.

## 1.4 Outline

The research project is organized into six chapters. Following the introductory chapter, Chapter 2 introduces the conceptual basis and technologies for indoor positioning, including signal-strength-based techniques, time- and angle-based techniques, deterministic ray tracing, and BIM-based environment modeling. Chapter 3 presents the proposed methodology, ensuring MOGA- optimization for AP deployment, and the evolution of the CNN-based multimodal fusion framework. Chapter 4 reports the implementation and experimental setup, defining the evaluation metrics used to assess localization and positioning performance. Chapter 5 presents the results of validation tests that evaluate the effectiveness of the proposed framework across various indoor environments. Finally, Chapter 6 concludes the project analysis and provides an outlook for future enhancements in robot localization.

# 2 State of the art

This section presents the theoretical foundations and recent surveys on indoor positioning for mobile robots. The research project covers sensing approaches, including radio-based techniques, the algorithms used to analyze positioning data, and the role of infrastructure design. Altogether, the elements provide a foundation for developing NGP systems that leverage the features introduced in IEEE 802.11 az.

## 2.1 Indoor positioning for mobile robots

Robots operating in indoor environments, such as factories, warehouses, offices, and hospitals, require accurate, robust, and scalable localization. Conventional localization techniques, such as inertial navigation and wheel odometry, provide short-term accuracy but eventually suffer from unbounded drift (a cumulative error over the long run). Vision or LiDAR-based SLAM systems overcome drift accumulation and used to construct a map of the surrounding environment while simultaneously estimating the pose of the robot. However, such localization techniques encounter significant challenges in indoor environments due to poor scene texture that help robust feature extraction, reduced LoS visibility, or varying lighting environments. Indoor characteristics, including glass partitions, corridor structures, and dynamic crowds, further hinder visual tracking. On the other hand, LiDAR-based techniques offer enhanced robustness to lighting variations but incur additional hardware costs and computational complexity (see table 1).

Wireless frameworks provide an appealing solution by extending building-scale coverage, offering low cost, and providing independence from surrounding light. Bluetooth low energy (BLE) and radio frequency identification (RFID) are power-efficient and widely available, although accuracy may be degraded. Ultra wideband (UWB) offers high accuracy through its wide bandwidth but requires specialized hardware. Wi-Fi is ubiquitous, practical, and easy to use, provided modern physical-layer characteristics are applied rather than just depending on signal strength. The enhancements in the IEEE 802.11 az standard promote decimeter-level accuracy, which works as the basic motivation for the positioning framework described in the following section.

## 2.2 Evolution of Wi-Fi technology

The development of Wi-Fi standards has significantly improved both spectral efficiency and channel observing capabilities. Succeeding Wi-Fi generations (802.11 n/ac/ax) established multiple input multiple output (MIMO) technology, multiple users-multiple inputs multiple outputs (MU-MIMO), orthogonal frequency division multiple access (OFDMA), and broadened channel bandwidths ranging from 40 MHz to 160 MHz, enhancing spatial and temporal resolution (see table 2).

Table 1: Indoor positioning technology landscape: comparison of onboard sensors vs. wireless infrastructure

<table>
<tr><th>Technology</th><th>Metrics</th><th>Advantages</th><th>Limitations</th></tr>
<tr><td colspan="4">Onboard sensors</td></tr>
<tr><td>Odometry + IMU<br>(Relative)</td><td>Acc: medium<br>Cost: low<br>Comp: low</td><td>+ Low-cost hardware<br>+ High update rate (100+ Hz)<br>+ No infrastructure needed<br>+ Simple implementation</td><td>– Unbounded drift over time<br>– Wheel slip errors<br>– No absolute positioning<br>– Requires periodic correction</td></tr>
<tr><td>Visual/LiDAR SLAM<br>(Absolute)</td><td>Acc: high<br>Cost: high<br>Comp: high</td><td>+ cm-level accuracy possible<br>+ Builds environmental maps<br>+ Loop closure reduces drift<br>+ Rich environmental data</td><td>– Lighting dependent (vision)<br>– Textureless areas problematic<br>– High computational cost<br>– Glass/crowds degrade tracking</td></tr>
<tr><td colspan="4">Wireless infrastructure</td></tr>
<tr><td>BLE<br>(RSSI-based)</td><td>Acc: 1–5 m<br>Cost: v. low<br>Comp: low</td><td>+ Very low power consumption<br>+ Ubiquitous (smartphones)<br>+ Low beacon cost<br>+ Easy deployment</td><td>– Low accuracy (1–5 m typical)<br>– High RSSI variance<br>– Human body shadowing<br>– Environmental sensitivity</td></tr>
<tr><td>UWB<br>(Time-based)</td><td>Acc:10–30 cm<br>Cost: high<br>Comp: medium</td><td>+ Very high accuracy (10–30 cm)<br>+ Wide bandwidth (>500 MHz)<br>+ Multipath resilient<br>+ Precise ToF measurements</td><td>– Requires dedicated hardware<br>– Limited device support<br>– Higher infrastructure cost<br>– Limited range (~50 m)</td></tr>
<tr><td>Wi-Fi 802.11 az<br>(Multi-modal)</td><td>Acc: 0.2–1 m<br>Cost: low<br>Comp: medium</td><td>+ Already deployed infrastructure<br>+ Sub-meter accuracy potential<br>+ Multi-modal (RSSI/ToF/AoA/ CSI)<br>+ Universal device support<br>+ Building-scale coverage</td><td>– Multipath in complex indoors<br>– NLOS bias in measurements<br>– Requires calibration<br>– AP placement optimization critical</td></tr>
</table>

*Metrics:* Acc = accuracy, Comp = complexity. Shaded row indicates focus of this study [2], [25].

802.11 mc introduced FTM for round trip time (RTT)-based ranging. However, ranging accuracy is limited by synchronization errors, NLOS transmission, and multi-path interference.

Table 2: Wi-Fi features across generations and the positioning impact [18]

| Standard | Key PHY/MAC features | Positioning relevance |
|---|---|---|
| 802.11 n | MIMO, 40 MHz | Basic multipath resolution |
| 802.11 ac | MU-MIMO, 80–160 MHz | Better delay/angle resolution |
| 802.11 ax | OFDMA, trigger-based | Coordinated exchanges |
| 802.11 mc | FTM/RTT | Ranging via RTT |
| **802.11 az** | 6 GHz, secure long training field (LTF)s | Robust ToF/AoA, richer CIRCSI/ |

IEEE 802.11 az demonstrates a fundamental change committed to positioning utility within Wi-Fi standards. The NGP presents key developments, trigger-based multi-user exchange approaches, secure LTFs, and optimized performance in the 6 GHz frequency band, with significantly enhanced effective bandwidths. The advancements enable more stable ToF measurements, enhanced AoA estimation using antenna arrays, and richer channel-

state information and CIR extraction, thereby improving indoor positioning accuracy. Theoretical experiments suggest that 802.11 az can achieve decimeter-level positioning accuracy under ideal SNR conditions and geometric conditions. However, achieving precision in practice requires optimizing the AP placement strategy and using robust signal processing to mitigate the effects of material interactions and dense multi-path scenarios. IEEE 802.11 az indicates a transition from basic communication-oriented Wi-Fi operation to a positioning-focused system design. For RTT ranging, previously IEEE 802.11 mc primarily promoted FTM. IEEE 802.11 az extends the physical-layer capabilities for trigger-based multi-user transmission and secure LTF designs.

ToF distance resolution relies on signal bandwidth and may be estimated as

$$\Delta d \approx \frac{c}{2B}$$

where $\Delta d$ represents distance resolution, $c$ represents the speed of light, and $B$ represents signal bandwidth. Finer delay resolution is achieved with higher bandwidth, and hence, compared to earlier Wi-Fi frameworks, ranging accuracy is enhanced. CSI quality and CIR estimation are improved by enhancing LTF design and multi-user sounding. Enhanced channel estimation promotes precise AoA computation utilizing antenna arrays and reliable ToF measurements. Under high SNR scenarios and acceptable geometric designs, stochastic assessment recommends decimeter-level positioning accuracy. Indoor regions, however, address material-based attenuation, NLOS transmission, delay spread, angular dispersion, and multi-path reflections. Reaching stochastic accuracy requires accurate AP deployment, advanced signal processing, and familiarity with environmental transmission modeling. The proposed framework integrates IEEE 802.11 az physical-layer measurements with BIM-based deterministic propagation modeling and GA optimization to handle intricate indoor transmission effects and enhance positioning accuracy. The following sections detail the requirements.

## 2.3 Radio wave characteristics for indoor positioning

Indoor positioning is influenced by multi-path and partial blockage. The most widely used positioning features and respective sub-categories are defined below (see Figure 1):

**Signal-based feature**, such as RSSI and CIR, is a simple positioning characteristic to achieve because RSSI is available on standard Wi-Fi hardware, and promotes rough, fingerprint-based positioning. Although RSSI readings are erratic due to human body interaction, environmental dynamics, and device orientation. Geometric information remains limited as RSSI demonstrates only received signal strength. Positions at similar or identical distances from the transmitter can produce identical RSSI measurements, enhancing geometric separation.

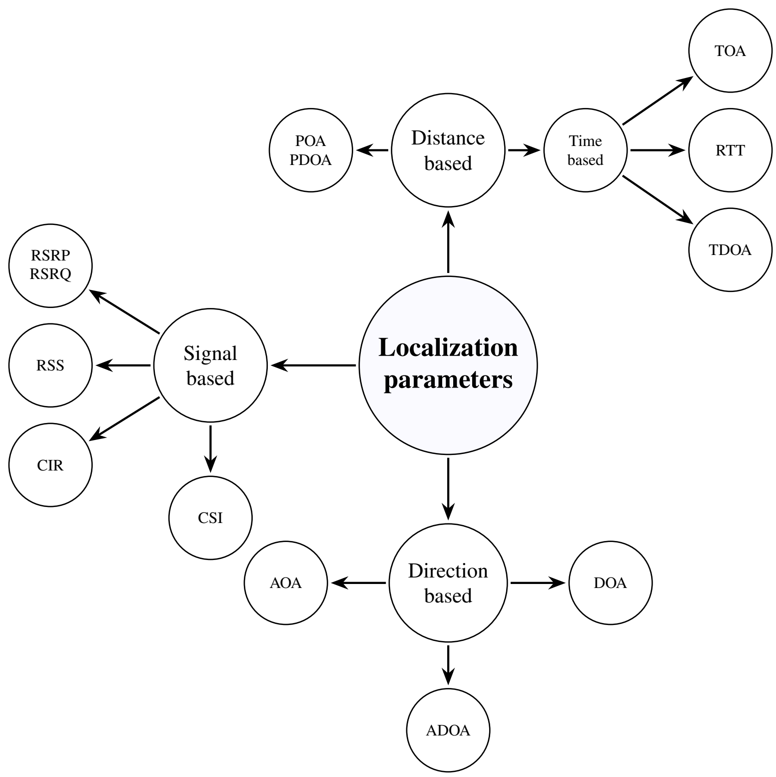


Figure 1: Radio wave technology

To eliminate such limitations, **time-based features**, including ToF. ToF measurements offer direct proxies for range calculation, with accuracy enhancements proportional to timing precision and signal bandwidth. The main issues include positive bias (Signal takes a detour through reflections, making objects appear farther away than actually the objects are) caused by NLOS conditions, processing delay calibration, and the need for accurate clock synchronization, as well as confusion between start and end multi-path components. Moreover, multi-path components can manipulate the start and end points, presenting more complexity.

To enhance the accuracy of range-based measurements, the following technique emphasizes **direction-based features**, including AoA. AoA offers directional information, which, when incorporated with range measurements, increases the capabilities of ToF. Hence, the whole system becomes accurate and reliable. AoA runs particularly well with multi-antenna arrays when the SNR is huge. The primary problems require accurate array calibration and the avoidance of inaccurate readings caused by multipath effects and sensitivity to factors such as polarization and antenna interaction. Such limitations can hinder the stability and accuracy of AoA measurements in particular scenarios.

Focus on enhancing the benefits of direction-based features **CIR** that allow the most detailed measurements, reflection patterns, analysis of signal delay, and, occasionally, phase information. The detailed information provided by CIR enables the fusion of ToF and AoA information in a unified system. But the measurements are very complex and large, susceptible to hardware offsets in the radio frequency (RF) chain, and require advanced modeling or training approaches, which make implementation difficult.

The additional strengths across various measurements suggest that combining multiple signal features is more reliable than relying on a single isolation feature. RSSI delivers energy-based information, ToF offers range, AoA gives direction, and CIR represents the most comprehensive measurements, detailed multi-path structure, encoding signal delay spread, mutually decreasing the limitations of every single technique. The described limitations highlight the requirement for the multi-modal fusion technique applied in the project.

## 2.4 Geometric, fingerprinting, and hybrid localization systems

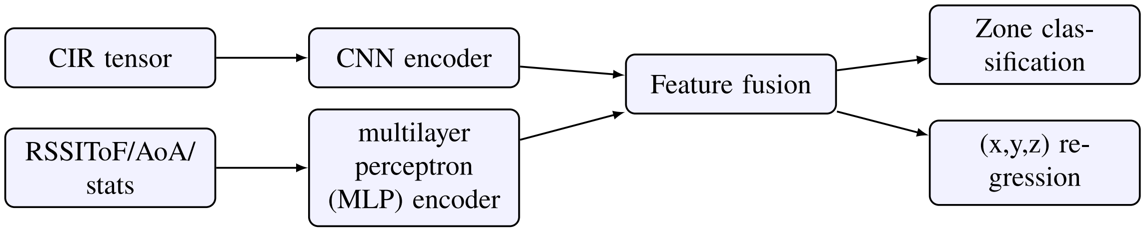


Figure 2: Intermediate fusion: CIR passes through a CNN encoder, statistics through a small MLP; the features are fused and fed to dual heads for robust zone classification and precise coordinates [25].

Indoor positioning algorithms can be classified into two main categories. **Geometric techniques** use triangulation, trilateration, or multilateration to convert angle measurements and distance into position estimates. Solutions may utilize nonlinear least-squares optimization or closed-form methods. Geometric techniques deliver mathematically clear and interpretable outcomes; however, such techniques may frequently fail under NLOS conditions and in an unfavorable AP framework. Reliability is typically communicated through GDOP metrics, with performance closely connected to AP deployment and LOS availability. On the contrary, **Fingerprinting or data-driven techniques** generate a mapping between measured radio frequency characteristics and represent zone or position labels (see Figure 2). Conventional techniques depend on k-nearest neighbor (K-nn) or support vector machines (SVM); however, modern techniques use deep learning, namely CNNs or transformer models, constantly treating CSI as 2-D characteristic map, analogous to images, to confirm pattern extraction. Fingerprinting naturally adapts to multi-path effects but still requires site-specific training information and thorough planning to guarantee generalization across environments.

The next sections demonstrate how training information is constructed and how the Wi-Fi positioning infrastructure is optimized.

## 2.5 BIM-integrated and deterministic ray tracing simulation

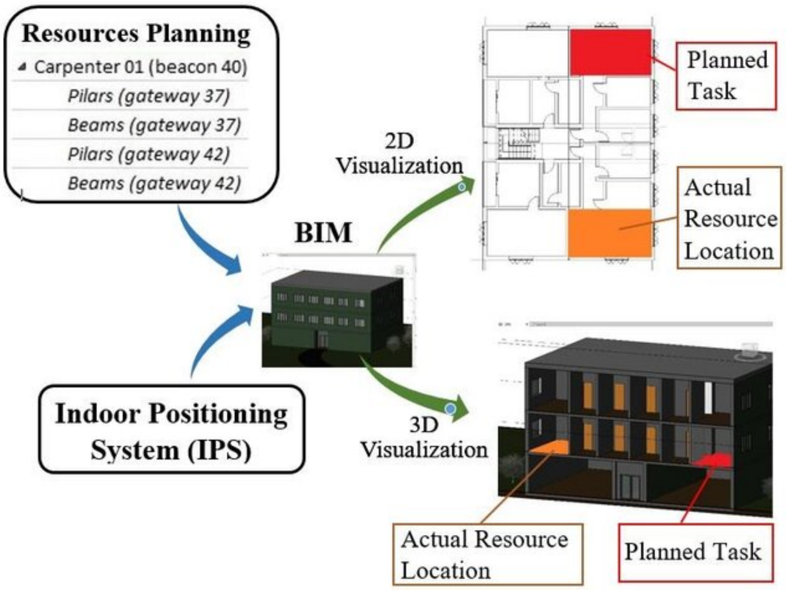


Figure 3: BIM in indoor positioning system [19]

BIM provides a complete 3-D representation of structural components, for instance, furniture, walls, glazing systems, partitions, and linked material components, including metal, concrete, wood, plasterboard, and glass characteristics. The theoretical combination of BIM information into the indoor positioning framework (see Figure 3). Fusing BIM geometric representations with a deterministic ray tracing approach enables the creation of a physically stable multi-path propagation pattern incorporating single and multiple-bounce reflections, LoS paths, associated delay characteristics, power characteristics, and transmission through materials.

Ray tracing analysis utilizing BIM information enables the generation of realistic channel impulse responses and the extraction of statistically accurate ToF and AoA distributions. The techniques prove important for fingerprint quality and geometric calculation performance, which are inherently tied to material components and building layout. Many materials exhibit unique electromagnetic properties, so metals generate strong reflections. Concrete delivers strong reflections with moderate fading. Plasterboard is highly sensitive to moisture and has low resilience.

Present ray tracing executions suffer certain limitations, such as approximations in small-scale structural effects, potentially incomplete material constant configurations, and a simplified view of diffuse scattering. Furthermore, computational costs increase significantly with the number of reflection orders selected. However, BIM-integrated ray tracing indicates the most practical technique for constructing site-specific, realistic propagation models at large-scale. The technique delivers training information that matches real-world scenarios (see Figure 4).

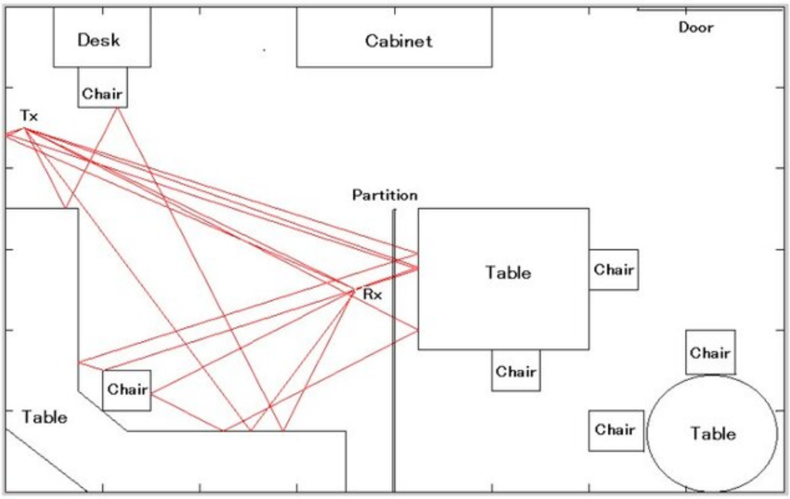


Figure 4: Ray tracing for indoor propagation modeling[11]

## 2.6 Optimization of AP deployment for indoor positioning applications

Conventional wireless local area network (WLAN) architecture focuses on enhancing coverage and data speed; however, positioning systems have distinct requirements. To obtain precise positioning, APs should be deployed in geometries that deliver low GDOP, good SNR coverage across the working environment, and wide-angle diversity, but also maintain the minimum number of APs to decrease cost.

The optimization operation selects the best subset of AP locations from a set of possible sites, which may be constrained by building limitations or grid layout. The choice should balance various goals, including providing sufficient coverage, maintaining good geometry for position evaluation, minimizing errors from NLOS conditions and deployment costs, and controlling the number of APs placed.

Finding the perfect combination is computationally expensive when many potential sites are available [13]. Alternatively, meta-heuristic techniques, such as GA, are also suitable for AP deployment, as the search space has a large number of possible AP combinations and contains several local optima. The GA-based method has possible deployments represented as binary strings, and the method evolves a population of user solutions using mutation, crossover, and selection. With a well-designed fitness function, GAs can effortlessly manage trade-offs between coverage, cost, and geometric quality. When penalties for the number of placed APs are considered, the GA-based method is capable of automatically selecting both the number of APs to employ and optimal deployment locations, based on positioning accuracy rather than only coverage. Therefore, GA-based methods form an important component of the suggested methodology.

# 3 Methodology

## 3.1 System overview

The proposed framework is planned as a synthesized four-stage design. The physical environment is optimized by deciding the most efficient AP positions by applying GA along with discrete optimization search space [5], [12]. Second, IEEE 802.11 az signals are simulated in a 3-D indoor representation to produce CIRs deduced from high efficiency long training fields (HE-LTF)[6]. Third, CIR data are integrated into fingerprint tensors. Finally, a deep CNN is trained for two tasks, such as continuous position estimation (regression) and zone classification (localization) [21].

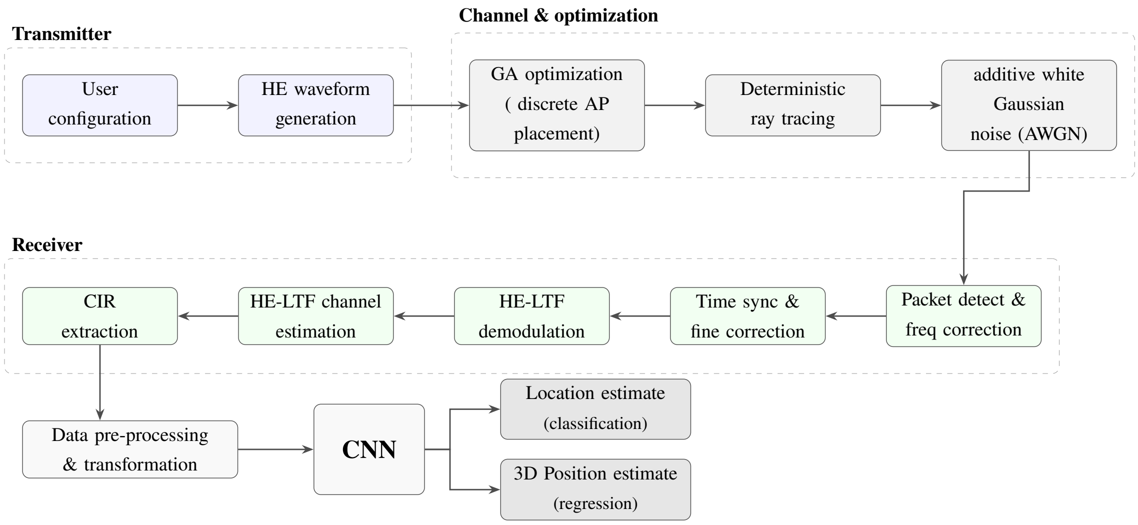


Figure 5: End-to-end pipeline

Signal processing stages and information path are described in the end-to-end pipeline, refer Figure 5. The transition from waveform generation at the transmitter side to dual-head result at the receiver side, going across the optimized deterministic channel, is demonstrated in the end-to-end pipeline.

Experiments include three channel bandwidths (CBW40, CBW80, CBW160), diverse antenna array sizes (linear 4 x 1, 2 x 1, 1 x 1), and two STA layouts (uniform and random grid). Utilizing IEEE 802.11 az leverages recent amendments that provide coarse positioning with high-precision capabilities and advanced channel estimation [17].

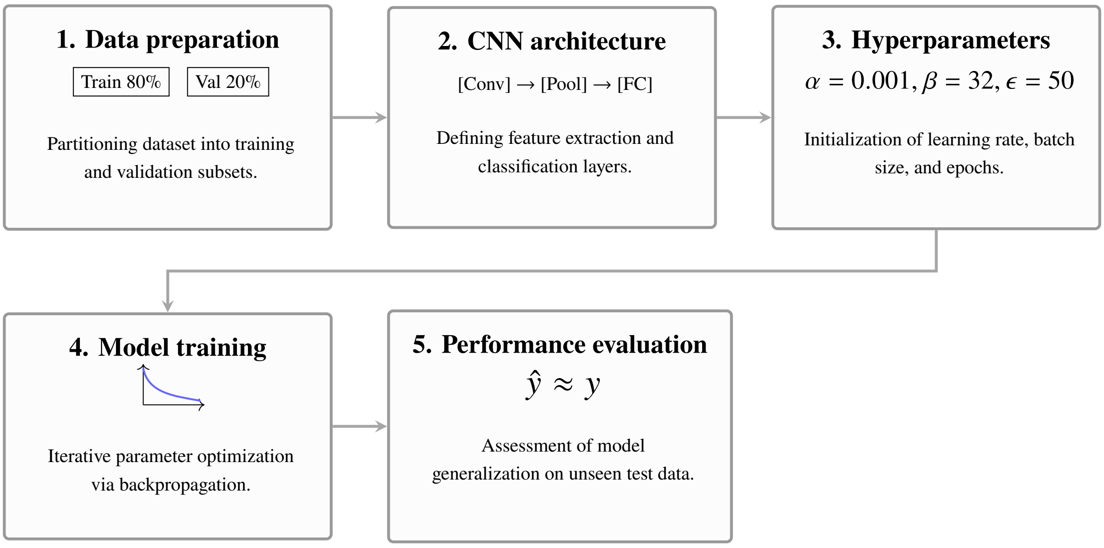


Figure 6: Workflow of the CNN implementation.

The implementation step, including data partitioning and repetitive parameter optimization, follows the particular logic detailed in the subsequent chapters, see Figure **??**.

## 3.2 GA-based AP placement

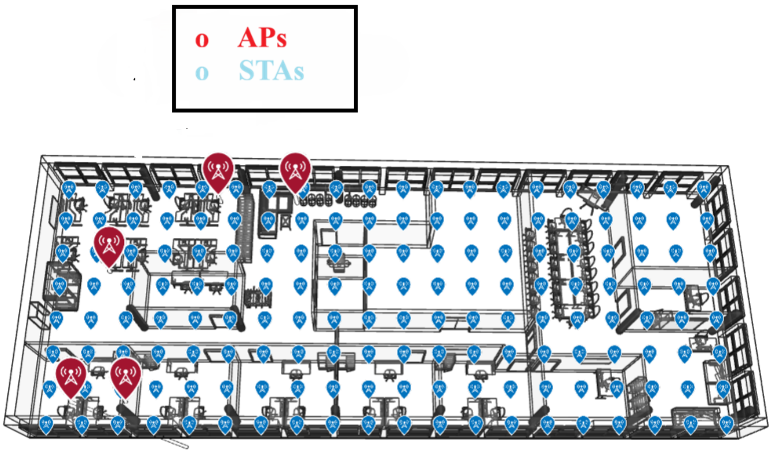


Figure 7: GA-based AP deployment, 0.7 STA separation

GAs are utilized as metaheuristic search methods to find near-optimal solutions in cluttered, multi-dimensional scenarios. GAs develop a population of candidate potential solutions using biological-inspired operators, namely mutation, crossover, and selection. Depending on the fitness function, each candidate outcome is computed to assess how well the fitness function serves the desired objective [5]. The algorithm continuously enhances the population by choosing the individuals with the maximum fitness to breed the future generation, see Figure 7.

A **single-objective GA** prioritizes enhancing an isolated scalar objective task. The optimization goal focused on lowering localization error or expanding AP coverage during the initial deployment stage [13]. Single-objective GAs perform well when the obstacle has a precisely defined optimization goal and is easy to interpret and execute. However, multiple contrasting goals should be considered simultaneously in real-life deployment conditions. Solutions are graded based on the Pareto front factor, where a solution is treated better only if the solution enhances one or more objectives without degrading others. The GA method generates a Pareto front that highlights the best trade-offs among competing objectives.

### 3.2.1 MOGA fitness formation for optimized AP deployment

AP deployment is modeled as a multi-objective optimization issue resolved by applying MOGA method. A deployment candidate is denoted by a binary decision vector

$$x \in \{0, 1\}^{N_c}$$

, where $N_c$ represents the overall number of candidate AP positions produced from the BIM design. The overall number of deployed APs is defined as:

$$N_{AP} = \sum_{a=1}^{N_c} I(x_a > 0.5)$$

### 3.2.2 Objective 1- coverage shortfall under K-coverage

Instead of single-link connectivity, precise 2-D positioning needs varied independent links. The minimal coverage requirement $K_{min} = 3$ is hence enforced. For every station s, The number of APs($n_s$) covering the station is defined as

$$n_s = \sum_{a \in S} \text{cover}(a, s)$$

where S represents the set of chosen APs and $n_s$ represents the number of clearly evident chosen APs at STA s.

The fractions of STAs fulfilling the K-coverage requirement is ($\rho$) defined as

$$\rho = \frac{1}{N_s} \sum_{s=1}^{N_s} \mathbf{1}(n_s \geq K_{\min})$$

The primary objective function ($f_1$) target is to reduce the coverage shortfall

$$f_1 = 1 - \rho$$

If $f_1 = 0$ denotes that all stations cover the three-link requirement.

### 3.2.3 Objective 2- Deployment cost

Wireless positioning infrastructure is represented via the total number of deployed APs ($f_2$)

$$f_2 = N_{\mathrm{AP}}$$

While preserving reasonable positioning performance, minimization of $f_2$ motivates complex deployments.

### 3.2.4 Objective 3- geometry quality utilizing a GDOP-like parameter

High coverage does not ensure accurate positioning. Poor spatial diversity among clearly evident APs raises the position ambiguity. Geometry standard is hence estimated utilizing a 3-D GDOP-like formation summarized from linear range-based positioning theory [16].

A huge penalty is allocated as the triangulation becomes unpredictable in the case where less than three APs are accessible.

#### 3.2.4.1 Significance of GDOP

GDOP addresses a parameter utilized to measure the effect of geometry on positioning accuracy. When distance measurements contain noise GDOP demonstrates how strongly AP geometry impact the final position error. Even with lower measurement noise clustered AP or linear deployments leads to bigger positional ambiguities. Adding GDOP parameter in MOGA objectives promotes AP designs that preserve larger angular separation around STAs. Spatial diversity decreases overlap between range circles, leads to reliable and distinct position estimates.

Unit vectors build the information matrix $G_s$, which represents the spatial geometry of the AP layout unaffected by RSSI. Amplification of measurement noise is reduced by minimizing the trace of the covariance matrix $Q_s$. Accurate AP configuration explicitly enhances positioning accuracy. Direction unit vectors from station position $p_s = [x_s, y_s]$ to AP positions $A_a = [x_a, y_a]$ are defined as $u_{a,s}$,

$$u_{a,s} = \frac{A_a - p_s}{\|A_a - p_s\|}$$

A geometry matrix is defined as $H_s$,

$$H_s = \begin{bmatrix} u_{x,1} & u_{y,1} & 1 \\ u_{x,2} & u_{y,2} & 1 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

The information matrix and covariance approximation are provided by $G_s$,

$$G_s = H_s^\top H_s, \quad Q_s = G_s^{-1}$$

The GDOP-like value for every STA s is defined as $g_s$,

$$g_s = \sqrt{\mathrm{trace}(Q_s)}$$

Smaller values represent better spatial diversity and enhanced positioning reliability. The ultimate geometry objective ($f_3$) is the average GDOP over all the stations

$$f_3 = \frac{1}{N_s} \sum_{s=1}^{N_s} g_s$$

### 3.2.5 AP-count restraint penalty

A quadratic penalty ($P$) is used if the number of APs exceeds the specified bounds ($minAP, maxAP$) to confirm realistic deployments.

$$P = 10^4 \max(0, \mathrm{minAP_{bound}} - N_{\mathrm{AP}})^2 + 10^4 \max(0, N_{\mathrm{AP}} - \mathrm{maxAP_{bound}})^2$$

The ultimate objective vector minimzed by MOGA method is hence defined as $F(x)$ [16],

$$F(x) = [f_1 + P, f_2 + P, f_3 + P]$$

### 3.2.6 Knee point-based decision making

A selection strategy identifies a singular optimal deployment from the Pareto front. The MOGA method produces a Pareto front of non-dominated solutions, i.e., optimal trade-offs, where minor refinements in one objective causes higher degradation in others. Estimating the Euclidean distance from every non-dominated solution to the ideal (but unrealistic) origin point (0,0,0) in objective space.

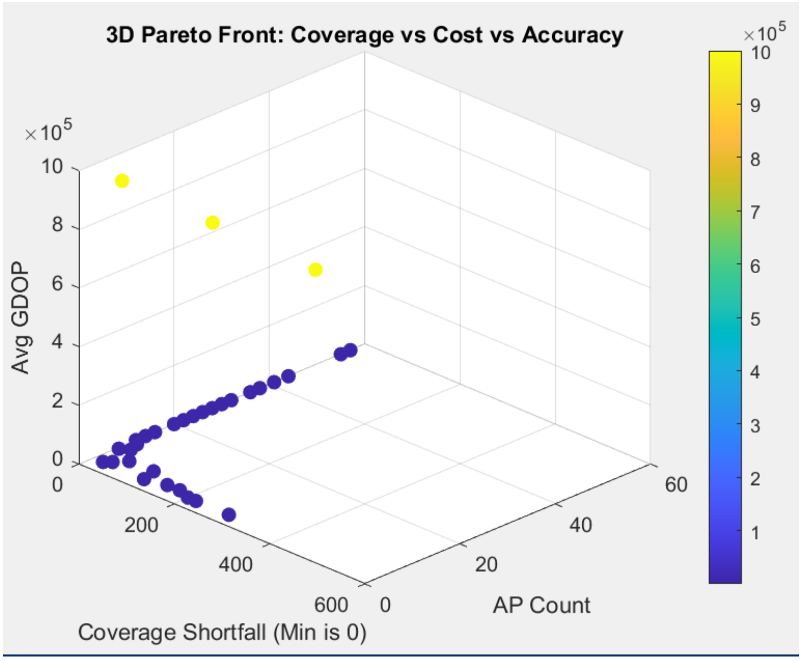


Figure 8: 3D Pareto front for AP deployment optimization. Dark blue points represent enhanced-accuracy configurations, whereas yellow points represent configurations penalized for failing to meet the K=3 constraint.

A knee-point represented in Figure 8, is a selection strategy [10], which is employed to select a balanced deployment that concurrently reduces coverage shortfall, deployment cost, and geometry dilution.

### 3.2.7 MOGA for AP deployment

To manage different deployment constraints the optimization process applies a MOGA framework. A Pareto-optimal set of solutions is generated by the optimization process to balance coverage, AP deployment cost, and geometry quality.

#### 3.2.7.1 Discrete optimization

To explore possible AP design layouts a discrete optimization strategy is applied. To form a potential deployment points the indoor floor area is divided into predefined candidate AP positions. The discretization transforms the infinite deployment search issue into a finite selection issue. Every design is encoded as a binary chromosome, where a value 1 demonstrates as an active AP at a grid location and 0 demonstrates no deployment at that location. The optimization process starts with an initial population of candidate layouts. Evolution covers mutation, crossover, and selection operations. Multi-objective selection leverages a Pareto-front method, where solutions are compared according to dominance over multiple objectives. To move toward effective network topologies, systematic estimation of candidate designs enables the non-dominated sorting genetic algorithm II (NSGA-II) [3].

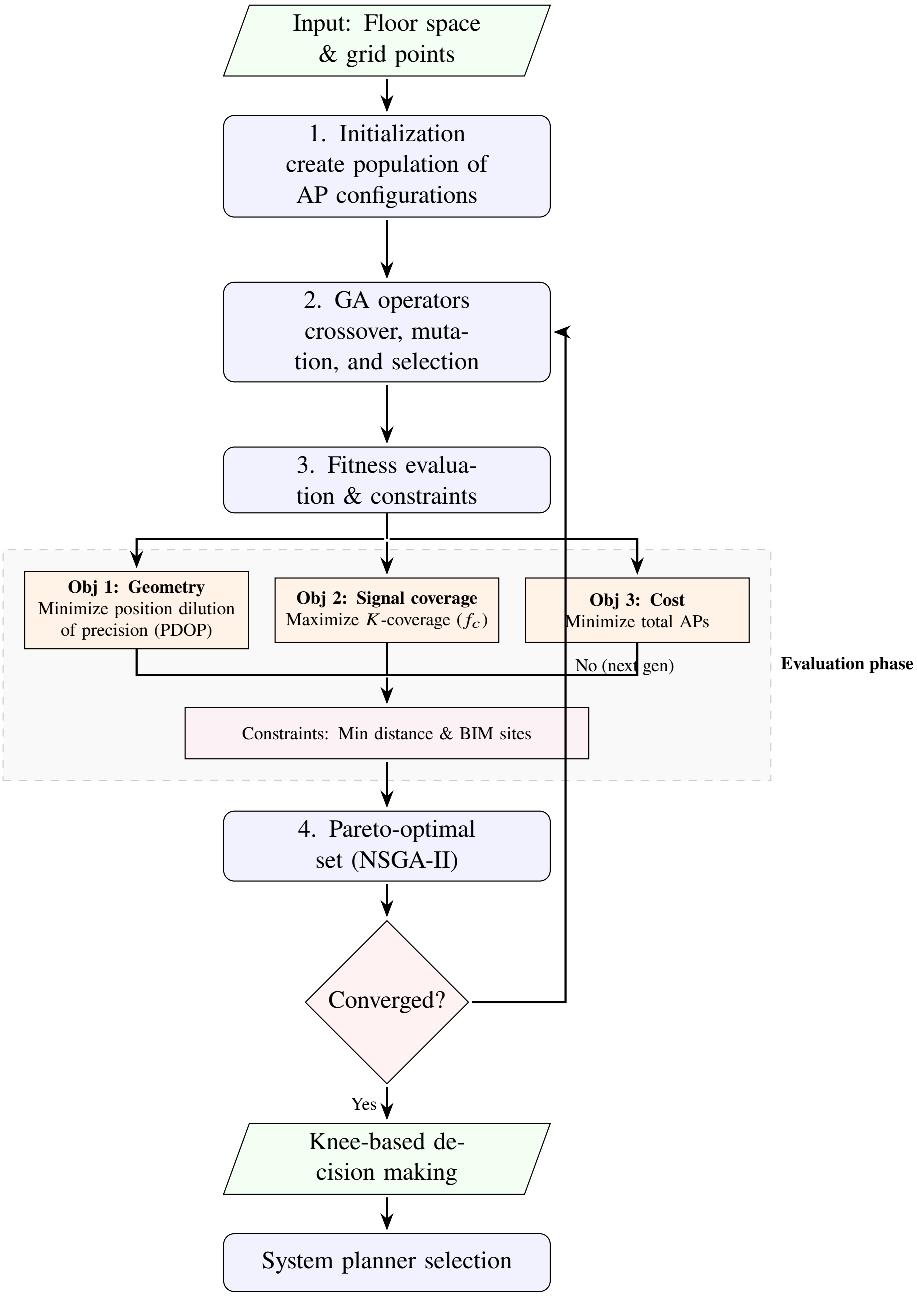


Figure 9: NSGA-II for Pareto-optimal AP placement with knee-based decision making.

A balanced deployment solution is determined by final design selection which applies a knee-point decision strategy.

### 3.2.8 Constraints and fitness evaluation

Alongside the three objectives, actual restrictions are employed to ensure AP deployments are achievable and well distributed. A minimal distance limit is enforced between any two APs, such that if two APs are deployed next to each other, a penalty is appended to the fitness solution. Complex deployments feel obstructed and propagate APs to cover separate zones. In the BIM model, user AP positions are not allowed at the approved installation sites, to avoid infeasible deployments. Each design is marked based on the three objectives during fitness evaluation. Employing normalization and a Pareto dominance check to compare solutions without falling apart into a single scalar, since two goals are to be reduced (error, cost) and increased (coverage), see Figure 9. To form the future generation, GA chooses the strongest design over a non-dominated sorting (e.g., NSGA-II style selection). The population unites towards better trade-offs among accuracy, coverage, and cost over successive generations. The result of the multi-objective GA is a Pareto-optimal set of AP deployment solutions. From the set, select a particular solution that matches the design requirement.

## 3.3 System modeling

The experimental environment is tailored to a realistic, simple indoor Institute of Digital and Autonomous Construction (IDAC) Technical University of Hamburg (TUHH) campus design represented as a 3-D mesh model stored in STL format [22]. The comprehensive 3-D model consists of walls, structural elements, and partitions, and is imported into MATLAB to serve as the simulation space. Within the framework, STAs and APs are placed to replicate different localization instances.

### 3.3.1 STA distribution strategy

Two spatial distribution strategies are used to reflect various application scenarios: **Uniform distribution** and **Random distribution**. **Uniform distribution** STAs are deployed on a uniform grid (e.g., 0.5 m spacing) across the indoors to verify spatial coverage. Allows for systematic examination of signal behavior and model performance at regularly spaced intervals. **Random distribution** STAs are randomly distributed to simulate real-world scenarios and erratic patterns. The variation establishes irregularity and helps test the robustness of the localization model under chaotic contexts.

### 3.3.2 Antenna array setup

The system promotes linear (M x 1) antenna array configurations for both the sender and receiver. The arrangements are used to assess the effect of antenna geometry and spatial diversity on localization performance and channel richness. By varying the array size, the behavior of the system can be analyzed across numerous MIMO scenarios.

### 3.3.3 Deterministic propagation modeling

Deterministic ray tracing is performed, see Figure 10 in Cartesian coordinates to simulate the channel propagation model [20]. The ray-tracing model accounts for key environmental factors, including glass, wood, concrete, brick, plasterboard, ceiling-board, marble, and

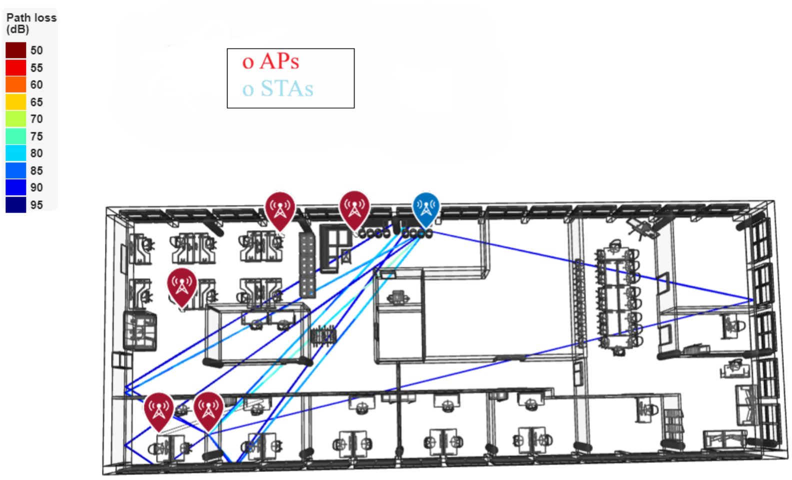


Figure 10: BIM floorplan and 3-Dvisualization illustrating ray-tracing paths (LoS and multi-path reflections) with color-coded AP and STA indicators.

polystyrene. Depending on the design complexity and surface material, signal propagation is tailored with up to five reflections. For each AP-STA pair, the ray tracing detects all feasible paths, such as LoS and multi-path factors, and evaluates corresponding attenuation and delays. The simulation correlates closely with the indoor propagation conditions defined by the IEEE 802.11 az standard.

## 3.4 CIR representation and dataset generation

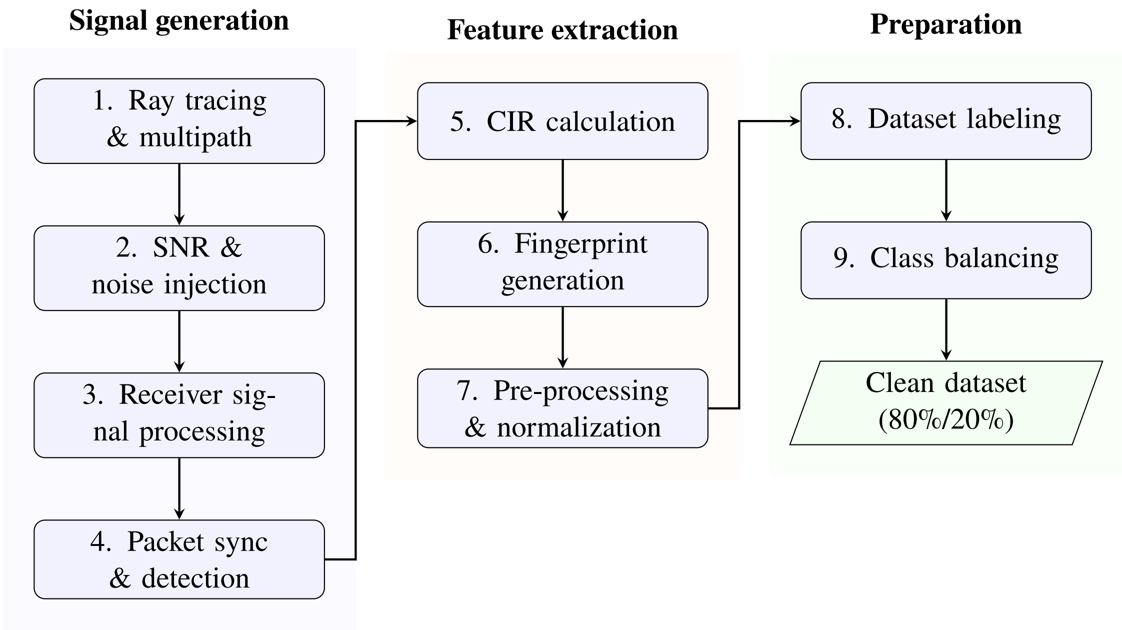


Figure 11: Structured workflow for IEEE 802.11 az CIR dataset generation.

The next section of the methodology is to creates a synthetic dataset of CIR fingerprints for training and evaluating the localization model, with AP placement calculated in the previous chapter. Figure 11 demonstrates the organized workflow, developing from primary signal generation to final dataset preparation. Data generation is based on the IEEE 802.11 az Wi-Fi standard, adding advanced physical-layer characteristics to confirm the fingerprint reflects actual indoor transmission. Each CIR measurement gets a label that matches the exact ground-truth location related to the transmitter-receive pair.

### 3.4.1 Simulation environment setup

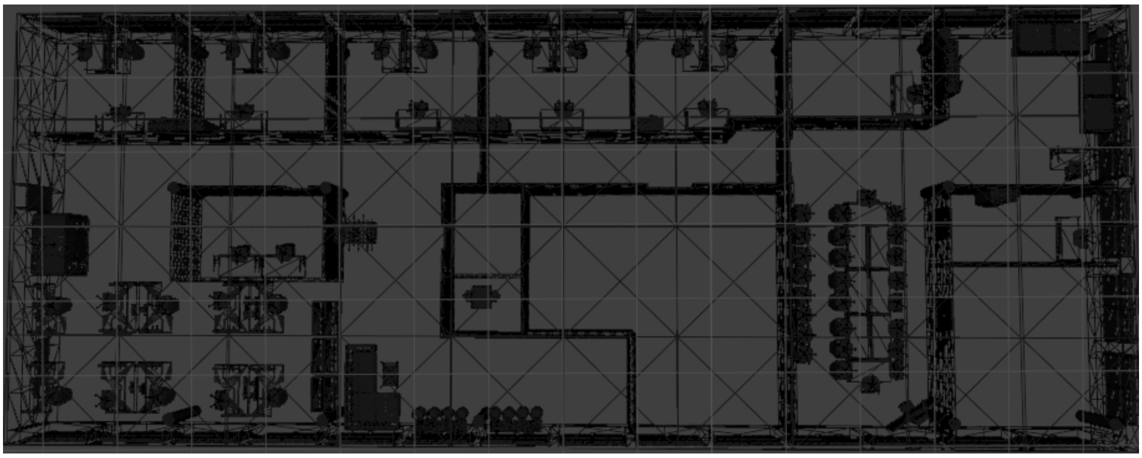

Figure 12: 3-D IDAC TUHH campus layout

To simulate signal propagation in a 3-D indoor environment, the ray-tracing technique is utilized and implemented in MATLAB (using the WLAN toolbox and related ray-tracing modules). BIM provides a detailed 3-D framework (imported from STL files of real floor design), see Figure 12, which indicates the geometry of floors, rooms, and walls, defined by the indoor space. The framework favors the propagation simulation to account for obstruction, reflections, and path loss due to the physical environment. The simulator receives an arrangement that determines the number of APs, the precise coordinates of each AP generated by the GA, and a compact set of receiver locations. The receiver locations, usually referred to as stations, are spread across the entire area of interest to cover connected parts of the indoor space. For creating fingerprint data, possible device locations are derived from station positions. Carrier frequency, for instance, the 5 GHz Wi-Fi band, and an appropriate MIMO antenna setup that meet IEEE 802.11 az requirements. The chosen parameters determine the resolution and richness of the resulting CIR measurements, with broader bandwidth and larger antenna arrays yielding finer time-domain and angle-domain resolution, respectively.

### 3.4.2 CIR generation process

CIR generation tracks a validated model framework in which IEEE 802.11 az high-efficiency waveforms are propagated from every AP and transmitted through the indoor space utilizing deterministic ray tracing. Inside the building, the ray-tracing model determines multi-path components generated by reflections and diffractions. Every receiver position accounts for a fused signal constructed by all multi-path achievements. To create multiple effective SNRs, AWGN is applied to the received waveform to simulate a range of realistic noise environments. Several SNR levels, starting from 0 dB to 40 dB with a step size of 2 dB, are used to create fingerprints covering both weak and strong-signal conditions, which are evaluated through multiple simulations. The resultant diversity in signal quality promotes the enhancements of a framework capable of managing a broad range of indoor propagation scenarios.

A standard 802.11 PHY receiver pipeline is executed to extract the CIR after signal transmission. To find the beginning of the packet, the receiver performs synchronization and packet detection, then applies fine-frequency offset correction and channel estimation methods as introduced in the 802.11 az protocol. The CIR between the AP and STA is calculated using the known training fields of the Wi-Fi packets (e.g, HE-LTF. For an AP-STA link, CIR basically records the channel's tap weights (delays of multi-path factors and amplitudes). Iterate the process for each combination of STA and AP, such that each station will have a unique CIR measurement from each AP in range. To form an isolated fingerprint for each position of the station, multi-AP CIR measurements are integrated. Each station generates $N$ CIR sequences, one from each AP-STA link, if $N$ APs are deployed. Every CIR sequence includes intricate channel coefficients corresponding to amplitudes and multi-path delays across time. The multiple CIR sequences can be structured either as individual channels in one data sample or fused into an isolated characteristic vector. At that particular position, the resulting fingerprint captures the spatial and multi-path characteristics of the radio environment.

### 3.4.3 Dataset composition and labeling

Every generated CIR fingerprint is matched with exact coordinates. Each sample supports dual localization tasks with two unique label types: a regression label and a classification label. The regression label provides continuous, precise coordinates within the building; on the other hand, the classification label identifies the discrete zone or specific room where the station is located. The room or zone label can be determined from the station coordinates by mapping to the floor space. To perform regression and classification, labels will be employed to train the deep learning model.

To ensure the dataset is appropriate, employ two essential pre-processing steps applied before training: normalization and class balancing. To support efficient neural network training, normalization is applied to the input characteristics. Because of device calibration or path loss, raw CIR measurements might vary in scale. For instance, applying mean subtraction and variance normalization, or by transforming values to a unit range, each fingerprint can be scaled. The scaling balances signal variations generated by hardware variations or signal degradation as the radio waves travel across the building. Consequently, all input characteristics exist in a similar empirical range. Normalization make sure that large CIR values do not suppress the model because of the magnitude variations instead of meaningful signal patterns. In the same way, according to building design, regression results such as exact coordinates can be scaled to a bounded interval, for instance, normalizing $x, y$ positions to the range [0,1]. During optimization, bounded coordinate values balance regression training and enhance empirical behavior. After applying normalization and other preprocessing steps, the process produces a clean, organized dataset of labeled CIR fingerprints suitable for both training and validation of deep learning models.

By applying class balancing, unequal sample distribution across location classes is managed. By balancing the training information so that every region is represented by approximately an equal number of samples, because some regions probably have more samples compared to others. Class balancing can be achieved by downweighting samples from classes with abundant information or by upweighting samples from classes with little information through augmentation or oversampling. Such balancing helps prevent a training model from biasing regions with a higher number of training samples. To ensure the dataset remains stable from the start, station points might be created in a ranked layout, such as allocating an equal number of points to each region.

## 3.5 Deep learning architecture for localization

The ultimate component of the methodology is a deep learning model that manages the refined CIR fingerprints as input and learns to predict the candidate location. The proposed framework is designed with two heads: a regression head to estimate precise coordinates and a classification head to estimate discrete locations. While expanding the model, start with a tailored shallow CNN as a reference and later transition to a more dense CNN (such as ResNet) to improve performance.

### 3.5.1 Baseline CNN model

The primary CNN applied was correspondingly shallow, comprising only a few convolutional layers, subsequently by fully connected layers. To establish a performance baseline, the model was used to verify that CIR information contains trainable characteristics for localization. The shallow CNN could absorb basic patterns in the CIR ( for instance, the absence or presence of a strong LoS or multi-path delay patterns) to differentiate among various locations. But, because of the bounded depth, the shallow network can eliminate cluttered structures in the data that are representative of fine-grained position differences. The baseline results suggested the need for a dense network capable of recording more fine-grained multi-path characteristics.

### 3.5.2 Advanced deep CNN (ResNet-based)

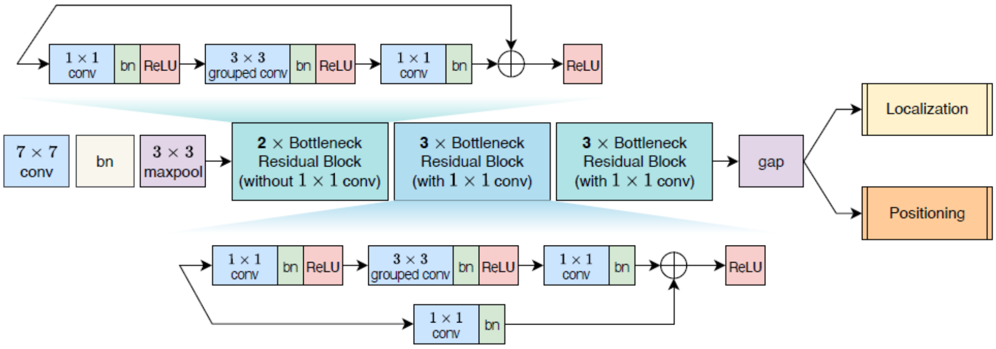


Figure 13: ResNet-18 [21]

Wireless signal-based indoor positioning requires extracting highly organized spatial features from CIR information. A deep CNN based on a modified residual network (ResNet)-18 framework is proposed, as in Figure 13, to model intricate multi-path patterns and spatial dependencies accurately. The residual framework improves feature representation quality, enhances computational efficiency, and secures gradient propagation.

The baseline ResNet-18 includes stacked residual blocks associated with skip connections. A residual block determines a mapping defined as

$$F_{\text{out}} = \mathcal{F}(F_{\text{in}}) + F_{\text{in}}$$

where $F(.)$ represents a residual transformation comprised of convolution, normalization, and activation layers, and $F_{in}$ represents the input feature tensor. The additional identity connections reduce degradation in dense networks and allow stable gradient flow throughout backpropagation.

But standard residual blocks include two successive 3x3 convolutional layers, which increase computational cost and parameter count. Bottleneck residual blocks are employed to enhance efficiency while preserving representational capacity.

#### 3.5.2.1 Bottleneck residual block formulation

By applying three convolutional layers, a bottleneck residual block presents expansion and dimensionality compression. The transformation can be defined as

$$F_{\text{out}} = C^{(3)}_{1\times1}\left(G_{3\times3}\left(C^{(1)}_{1\times1}(F_{\text{in}})\right)\right) + C^{(s)}_{1\times1}(F_{\text{in}})$$

where $C^1_{1*1}$ represents the regular convolution, $G_{3*3}$ represents the grouped convolution, $C^3_{1*1}$ restores channel dimensionality, and $C^s_{1*1}$ represents a projection layer used only when input and output dimensions vary.

Let the input tensor have dimensions $c_{in}$. The primary 1 x 1 convolution lowers the channel dimension to $c_{mid}$, where $c_{mid} < c_{in}$. The secondary grouped convolution refines spatial features inside the reduced channel space. The ultimate 1x1 convolution increases the number of channels to $c_{out}$ while maintaining network depth and representational strength; the bottleneck framework lowers the parameter count. During training, the residual total confirms gradient stability.

#### 3.5.2.2 Grouped convolution formulation

Grouped convolution separates input channels into $n_G$ groups. Applying separate filter banks, every group is convolved individually. Let $c_{in}$ represent the number of input channels, $c_{out}$ represent the number of output channels, $m_H * m_W$ represent the filter dimensions, and $n_G$ represent the number of groups.

The overall number of trainable parameters in a grouped convolutional layer without bias is

$$N_{\text{params}} = \frac{c_{\text{in}} \cdot c_{\text{out}} \cdot m_H \cdot m_W}{n_G}$$

where $n_G = 1$, the execution lowers to standard convolution. When $n_G = c_{in}$, the execution becomes depthwise convolution.

While preserving the potential to learn different spatial patterns across feature partitions, grouped convolution incurs roughly proportional performance costs as $n_G$ increases. Grouped convolution enhances efficiency without corrupting spatial feature extraction quality for wireless localization tasks.

# 4 Implementation

Following the system framework illustrated in Chapter 3, this chapter demonstrates the actual execution of each component of the proposed framework. The execution transforms the theoretical contexts into a replicable MATLAB-based simulation pipeline. The pipeline consists of enhanced AP deployment, ray tracing based on indoor channel modeling, IEEE 802.11 az characteristic generation, and deep learning for indoor positioning. Every step is carried out systematically to confirm the reliability of the methodological concept with respect to the simulated data.

## 4.1 Creating an indoor environment

The main objective is to achieve centimeter-level accuracy using the 802.11 az standard. The complete simulation and optimization process is managed utilizing **MATLAB R2023b** using specialized toolboxes that include **phased array system toolbox, signal processing toolbox, communication toolbox, optimization toolbox, global optimization toolbox, statistics and machine learning toolbox, WLAN toolbox, and deep-learning toolbox**. Toolboxes provide the necessary functions utilized to simulate wireless scenarios, extract features, and train machine learning models.

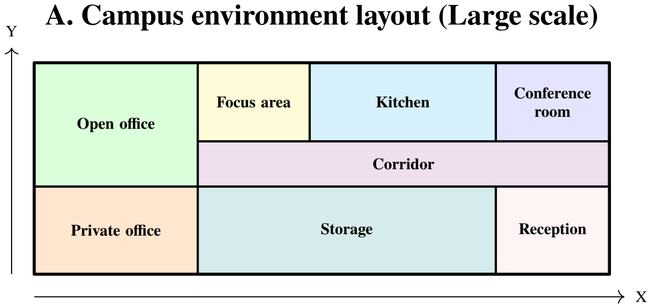


Figure 14: simulation environments: large-scale campus layout

### 4.1.1 Hardware and software setup

Each simulation and model training are performed on a system equipped with an Intel Core i7-1165G7 processor (11th generation, 2.8 GHz), 16GB of RAM, Intel Iris Xe graphics, and 477 GB of storage. The configuration is adequate for managing computationally intensive tasks such as ray tracing, CIR extraction, and deep neural network training. In the project, simulation scenarios is conducted utilizing **Blender** to design a 3-D framework, saved as an .STL file. The file represents the spatial foundation for determining the positions of STAs and APs. The design agrees to test various AP configurations and optimize AP deployment for higher coverage. GA are employed to optimize the geometric distribution of APs within the room, coverage, and cost.

The STL- generated 3-D model provides a practical spatial environment that simulates the transmission characteristics of various materials in the room, including **concrete, wood, metal, glass, and plasterboard**. Such materials exhibit numerous fluctuating signal losses, which affect the coverage area of the APs. The setup closely matches real-world environments and accounts for the actual assessment of AP deployment algorithms.

The computing paradigm utilized for the simulations, a Laptop with Intel processor, helps with multi-threaded operations, which are important for managing huge datasets and simultaneous calculations effortlessly. The paradigm effortlessly manages **CIR extraction and ray tracing** and performs the data augmentation steps, which require substantial computational power.

## 4.2 Simulation environment and data generation

The primary objective of the simulation is to **optimize AP deployment** within a room to increase coverage, while accounting for factors such as geometric distribution and cost. STA positions are first produced employing two distribution techniques that include **uniform and random**. The separation between the STA is changed in the simulation, with standard distances such as **0.7 meters, and 0.5 meters** used across various environments. The separation distances are selected to examine how adaptation in station diversity and spatial distribution manipulate access point deployment and coverage performance. The proposed technique encourages understanding coverage dynamics across different indoor environments.

Now that the STAs are determined, the environment setup is finished by using the ray tracing model to simulate the transmission of signals between STAs and APs. Signal transmission within the environment is simulated using a ray-tracing-based transmission framework that accounts for diffraction, reflection, and electromagnetic wave propagation based on the material properties of the zone or room. The framework is important for precisely estimating the coverage area and signal strength of the APs.

The number of STAs is defined by the STA separation distance and the distribution technique when the focus is on data generation. In uniform distribution, the number of STAs is evaluated as the ceiling of the room divided by the STA separation distance, confirming that the STAs are uniformly distributed across the room.

$$N_{STAs} = \prod_{i \in \{x,y,z\}} \left\lceil \frac{\Delta_i}{s} \right\rceil = \left\lceil \frac{L}{s} \right\rceil \cdot \left\lceil \frac{W}{s} \right\rceil \cdot \left\lceil \frac{H}{s} \right\rceil \tag{1}$$

Where L,W, and H is length, width, and height of the room, respectively. Separation distance (e.g, 0.5m) is s, and $N_{STAs}$ represents the count of STAs.

On the contrary, random distribution presents a scaling parameter to accommodate the number of STAs on the basis of separation distance, allowing for compact or deeper distributions which relies on the requirements of the simulation.

$$N_{STAs} = \min\left(1000, \left\lfloor N_{base} \cdot \left(\frac{s_{ref}}{s}\right)^2 \right\rceil\right) \tag{2}$$

where, $N_{STAs}$ is the final number of random stations generated.$N_{base} = 1000$ is your starting population constant.$s_{ref} = 0.5$ is your reference separation distance. $s$ is your current separation distance (STASep). The term $\left(\frac{0.5}{s}\right)^2$ is the quadratic scale factor.$\lfloor \dots \rceil$ represents the rounding to the nearest integer. To handle resource optimization, the total number of STAs is capped at 1000.

### 4.2.1 Signal standard and feature generation

Now that ray tracing is completed, IEEE 802.11 az standard ranging waveforms are produced by applying the heRangingConfig function. The ray-traced multi-path information, which uses the 802.11 az physical-layer signal framework across various SNR levels to create labeled CIR-based feature tensors.

The features basically consist of the CIR, enhanced by multi-modal information, namely RSSI, AoA, and ToF. The labels include the exact coordinates for regression and categorical room identifiers for classification. Lastly, noise is added to simulate random jitter in the STA location. The noise models capture the irregularity in actual calculations, where STA locations may not be precisely aligned with the ideal grid due to environmental factors, such as signal distortion or physical blockage.

## 4.3 STL-based coordinate restrictions

Physical building boundaries should be followed by candidate AP positions. Spatial limits are retrieved explicitly from STL models so that discrete grid points exist in the structural box.

### 4.3.1 Boundary extraction and definition of bounding box

The minimum and maximum spatial extents along the X, Y, and Z axes is calculated by parsing STL vertices. Such limits describe the Bounding Box ($B_{\text{box}}$), which addresses as the global search space for optimization. Derived from building information, position bounds are determined as X-axis: [-25.5447, 13.5017], Y-axis: [62.7973, 78.2469], Z-axis (floor level): [25.0903]. To avoid AP deployment in non-physical space any candidate point outside the bounds is instantly declined.

### 4.3.2 Polyshape validation and discrete grid creation

Inside the defined $B_{\text{box}}$, a discrete grid of possible APs positions. An added polyshape validation step is needed as the $B_{\text{box}}$ forms a simple rectangle and building design can be erratic.

A 2-D estimation of the STL model generates a simple polyshape representation of the building design. A geometric point-in-polygon test estimates at every grid point ($P_{\text{grid}}$). Chromosome encoding consists isolated points located within the building design

$$P_{\text{grid}} \in \text{Interior(Building Polyshape)}$$

### 4.3.3 Antenna height and vertical offset

Actual signal transmission is ensured by fixed vertical offsets. STA receivers are placed at a typical height $H_{sta}$, while APs are placed at a height $H_{AP}$

$$Z_{\text{AP}} = Z_{\text{floor}} + H_{\text{AP}}$$

$$Z_{\text{STA}} = Z_{\text{floor}} + H_{\text{STA}}$$

Enabling $H_{AP}$ to 2.2 m above the floor space preserves LoS scenarios over little obstacles and validate for signal degradation resulting from plasterboard ceiling structures described in the BIM framework.

## 4.4 Feature extraction and fusion

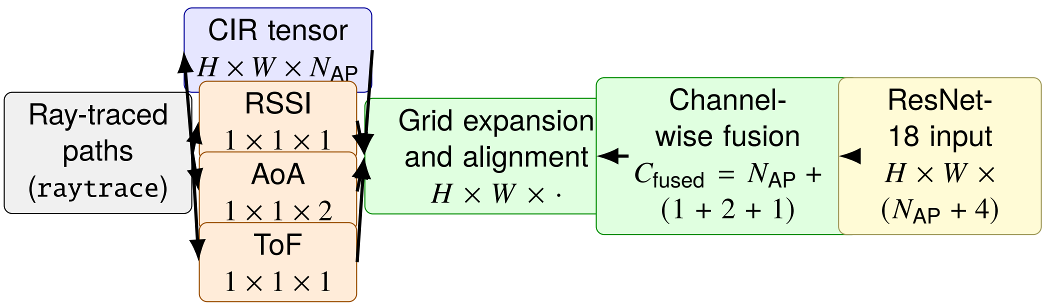


Figure 15: Multi-modal feature extraction and fusion pipeline (implementation-consistent).

The proposed framework includes an organized multi-modal characteristic extraction and fusion strategy designed to combine various signal characteristics into a unified radio fingerprint. The complete pipeline is demonstrated in Figure 15.

### 4.4.1 STA-based sample mapping

Construction of a physical reference map demands precise 3-D positions for each station in the environment. Spatial alignment involves associating each signal measurement with the nearest physical station position by minimizing Euclidean distance.

$$s = \arg\min_{j} \sqrt{(x_n - x_j)^2 + (y_n - y_j)^2 + (z_n - z_j)^2} \tag{3}$$

Here, $(x_n, y_n, z_n)$ represents the coordinate of sample $n$- and $(x_j, y_j, z_j)$ represents the coordinate of the station $j$. The mathematical operation confirms every sample is correctly connected to the physical design of the room.

### 4.4.2 CIR tensor construction

Signal propagation evaluation follows sample mapping to explore propagation rays from each AP to each STA. The dominant path represents to the trajectory with higher receiver power and lower path loss. CIR generation extracts the amplitude and time-delay characteristics linked with the chosen propagation paths.

In the implementation, the CIR tensor is determined as

$$\mathbf{F}_{\mathrm{CIR}} \in \mathbb{R}^{H \times W \times N_{\mathrm{AP}} \times N}$$

where $H$ is the number of stored delay bins, $W$ corresponds to the integrated spatial dimension formed by space-time streams and receive antenna components, $N_{\mathrm{AP}}$ is the number of deployed APs, $N$ is the total number of samples( number of STA x number of SNRlevels in dB).

Here, AP particular CIR feature maps are stacked by the third dimension. Allowing the model to learn spatial diversity over numerous infrastructure nodes, as every channel represents one AP viewpoint.

### 4.4.3 Feature extraction of dominant path

Ensuring that the RSSI uses the most stable signal information at each position by emphasizing the strongest path helps reduce distortion that might degrade the signal.

$$RSSI_s = P_{Tx} - \min(\mathcal{L}_{path,s}) \tag{4}$$

Extracting the ToF is achieved from the equivalent propagation delay, whereas AoA is achieved from azimuth (horizontal) and elevation (vertical) angles. Both ToF and AoA parameters are calculated per station and describe dominant large-scale channel characteristic behavior.

### 4.4.4 Alignment of tensor and grid expansion

While AoA is a 2-D vector (azimuth, elevation), ToF and RSSI are scalar values. To allow fusion with the spatial tensor, all three characteristics are expanded beyond the $H \times W$ grid

$$\mathbf{X}_{RSSI} \in \mathbb{R}^{H \times W \times 1 \times N}$$

$$\mathbf{X}_{AoA} \in \mathbb{R}^{H \times W \times 2 \times N}$$

$$\mathbf{X}_{ToF} \in \mathbb{R}^{H \times W \times 1 \times N}$$

In Figure 15, the notation $H \times W \times \cdot$ indicates that the channel dimension depends on the number of feature maps prior to concatenation.

### 4.4.5 Channel-wise fusion

Feature fusion is executed by integrating along the channel axis

$$\mathbf{F}_{\text{fused}} = [\mathbf{F}_{\text{CIR}} \parallel \mathbf{F}_{\text{RSSI}} \parallel \mathbf{F}_{\text{AoA}} \parallel \mathbf{F}_{\text{ToF}}] \tag{5}$$

where $\parallel$ denotes the concatenation operator.

Now RSSI and ToF provide one channel, whereas AoA provides two channels. Hence, the ultimate channel dimension becomes

$$C_{\text{fused}} = N_{\text{AP}} + 4$$

For example, with $N_{\text{AP}} = 14$, the tensor size changes from

$$[H, W, 14, N] \to [H, W, 18, N].$$

The channel-wise fusion allows the convolutional network to learn mutually CIR provides fine-grained multi-path architecture, RSSI provides large-scale power variations, AoA provides geometric directionality, and ToF provides range estimation information.

The ResNet-18 localization framework uses the fused tensor as input.

### 4.4.6 Temporal resolution and selection of delay bins

Precise device localization relies on capturing high-resolution radio fingerprints using CSI. Multi-path reflections across discrete time intervals are represented by CIR, identical to how acoustic echoes illustrate sound reflections. Temporal resolution is determined by signal bandwidth. Larger bandwidth allows higher-resolution multi-path component separation, and hence, spatial discrimination is improved. The delay dimension $H$ represents to the number of stored samples from the cyclic prefix (CP) duration of the OFDM symbol. Inter-symbol interference is mitigated by CP by accommodating delayed multi-path components.

The number of delay bins is, hence, calculated as

$$H = 0.75 \times CP_{\text{Length}} \tag{6}$$

where $CP_{Length}$ in samples relies on both the chosen guard interval and the sampling rate linked with the chosen bandwidth. An OFDM symbol offset factor of 0.75 is applied to preserve exclusively the most reliable 75% of the cyclic prefix duration. Moreover, efficiently removing noise and late-arriving reflections might reduce positioning accuracy.

The CP length in samples increases proportionally with the sampling rate for a fixed guard interval (in the project default = 1.6 µs) and, hence, with the channel bandwidth. As a result, a large number of delay bins and finer temporal resolution of the CIR is provided by higher bandwidth.

Table 3: Relationship between bandwidth, cyclic prefix length, and resultant measurement points ($H$). Using the default guard interval 1.6 µs

| **Wi-Fi signal width (Bandwidth)** | **Total samples** ($CP_{Length}$) | **Final points** ($H = 0.75 \times CP_{Length}$) | **Detail level** |
|---|---|---|---|
| 40 MHz | 64 | 48 | Better |
| 80 MHz | 128 | 96 | Good |
| 160 MHz | 256 | 192 | Excellent |

See Table 3 outlines the relationship between bandwidth, CP length, and stored delay bins $H$.

Higher bandwidth corresponds to a higher density of measurement points, yielding a comprehensive spatial map for the ResNet-18 model.

With the rich multi-modal input, the ResNet-18 model can learn from both the temporal architecture of the CIR and the geometric constraints utilized to enhance zone classification, given by RSSI, ToF, and AoA information.

## 4.5 Pre-processing and augmentation

To develop a deep learning dataset in MATLAB, data processing begins with normalization. For classification tasks, room labels are changed into categorical values. All samples identified as class 0 are removed during data processing. Such samples match positions where ray tracing did not satisfy the coverage prerequisite. Maintaining such samples would lead to invalid signal information and adversely affect classification accuracy. Therefore, eliminating class 0 confirms that only physically useful and stable information is employed for training.

Next, the focus is on class imbalance after cleaning the dataset. Imbalance usually occurs because rooms differ in size and number of stations, resulting in uneven sample counts per room. To overcome, a median-based balancing technique is used. Primarily, the median sample count across all valid rooms is used as the target value. Rooms with more samples than the median are randomly undersampled; however, rooms with fewer samples are increased by reconstructing existing samples. The balancing step prevents the ResNet-18 framework from promoting larger rooms and helps balance training across indoor rooms.

To further enhance framework robustness, data augmentation is applied to the balanced dataset. According to a predefined augmentation factor, every primary sample is replicated. Two transformations are applied for every replica. To start with, the CIR is circularly shifted along the time axis, simulating minimal changes in signal delay that are common in actual wireless hardware. Second, random Gaussian noise is introduced to the signal. The noise demonstrates measurement unpredictability and environmental distortion. Integration of the transformations prevents the framework from learning the same signal patterns and encourages the framework to learn normal features that hold firm in real-world situations.

Together, the preprocessing steps verify that the final dataset is robust, measurable, and balanced. The preparation enables the deep learning framework to generalize well to real indoor settings. Simultaneously, balancing and augmentation confirm that the dataset is sufficiently large for efficient deep learning and diverse enough to reflect real-world scenarios. The preparation step fills the gap between a managed simulation scenario and an actual indoor positioning system equipped with robust coordinate regression and zone-level classification.

## 4.6 Spatial data partitioning and model adaptation

After data augmentation, the final dataset is partitioned utilizing a split-by-STA strategy. Every fingerprint is linked with the closest physical station by applying Euclidean distance minimization [7]. The dataset is then split, with 20% of the physical station identifiers reserved for the validation set. Evaluation of the framework is conducted on spatial positions unavailable during the training phase, making the evaluation a difficult test of the capability to generalize across indoor scenarios.

To confirm that small structural rooms, such as isolated desks, are regularly demonstrated throughout framework estimation, a stratified spatial split [15] is applied. The dataset is split into 20% validation for each room category, rather than a global random partition. Stratified splitting confirms that even rooms with a little station density are present in both the training and validation sets.

## 4.7 Multi-task output design

### 4.7.1 Training procedure and hyper-parameters

Using the synthesized CIR dataset mentioned earlier need to train the deep learning model. All neural networks are implemented on a separate CPU employing the deep learning toolbox of MATLAB. To enable model tuning and prevent overfitting, split the dataset into training and validation sets (e.g., 80/20). The training procedure leverages mini-batch gradient descent. In the experiments, a comparatively small batch size (e.g., 32 samples per batch) is used to balance the gradient estimates, along with the Adam optimizer with an initial learning rate of $1e-4$. To lower the combined loss, each training epoch iterates over all training samples in batches and updates the model weights. Until the validation loss stops improving, or train for an adequate number of epochs (order of 3). Analyze the performance of the model on the validation set throughout training. The validation regression error shows the average localization error in distance, and the classification accuracy on validation data indicates how well the model correctly identifies the zone of the candidate, as illustrated in table 4. The metrics advise on hyperparameter tuning (e.g., if the model is overfitting, the tuning process can be terminated early; if underfitting, more training epochs or increased capacity may be needed).

Leveraging typical regularization techniques to enhance generalization, namely $L_2$ weight regularization, dropout in the fully-connected layers, or early termination based on validation performance. To make the model more robust to measurement modifications, data augmentation may be applied by introducing slight changes to the CIR data (for instance, adding minimal noise or random phase shifts).

Table 4: Performance metrics used for classification and localization evaluation

| **Task** | **Metric** | **Mathematical definition** |
|---|---|---|
| Classification | Accuracy | $\text{Accuracy} = \frac{\sum_{i=1}^{K} C_{ii}}{\sum_{i=1}^{K}\sum_{j=1}^{K} C_{ij}}$ |
| Regression | Positioning Error | $e_n = \sqrt{(x_n - \hat{x}_n)^2 + (y_n - \hat{y}_n)^2 + (z_n - \hat{z}_n)^2}$ |
| Regression | RMSE | $\text{RMSE} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} e_n^2}$ |

The tailored ResNet-18 framework is augmented with a dual-head output architecture to jointly perform regression and classification. The multi-task learning (MTL) technique allows the framework to share minor feature demonstration, where the shared convolutional layers extract basic signal patterns that is beneficial for both exact position estimate and room classification.

### 4.7.2 Regression head

Fine-grained spatial estimation is implemented by the regression head. The head uses a linear activation function on the output, a 3-D vector, showing the (x, y, z) positions. The regression head uses a mean squared error (MSE) loss function to lower the Euclidean distance between the estimated and actual positions. The addition of the regression head forces the network to preserve high-frequency spatial details that can be absent in a pure classification framework.

### 4.7.3 Classification head

The classification head is responsible for zone-level recognition. Classification involves a fully connected layer corresponding to the number of acceptable room categories (e.g., conference rooms, offices, desks with chairs). The classification categories match directly to the physical rooms demonstrated in the IDAC campus environment design indicated in figure 14. The indoor scenario is divided into eight different areas: the conference room, open office, storage or utility, private office, small meeting or focus area, kitchen or break area, corridor or main passage, and reception entry zone . Each coordinate sample (x, y, z) located within a colored area in the plan view receives an actual label corresponding to the name of that room. The categorical mapping confirms that the spatial distribution of the generated dataset aligns exactly with the structural layout of the room.

APs are strategically classified inside the rooms to record the distinct multi-path fingerprint feature of every room. The framework applies special signal features to distinguish between nearby rooms, such as the transition between kitchen and conference room, which might share the same power levels.

The fusion of a multi-modal framework improves the capability of the model to differentiate between rooms that has shortage of physical partitions. Integrating AoA, including both azimuth and elevation elements, with ToF allows the network to recognize transparent boundaries even in open-space region. The specialized combination is specifically efficiently during the period of transition between the kitchen and the conference room, where angular and temporal angles give the important geometric restrictions for higher accuracy region classification.

A **softmax activation** function is implemented to the output, converting the raw network counts into a probability distribution. During training, the classification task is improved by applying a **cross-entropy** function, which penalizes the difference between the expected room probabilities and the actual categorical labels.

### 4.7.4 Combined loss function

To learn the network in a fused fashion, the two isolated losses are integrated into an individual objective function

$$L_{total} = L_{class} + \lambda L_{reg} \tag{7}$$

where $\lambda$ is a hyperparameter that stabilizes the contribution of positioning error beside the classification accuracy. The combined optimization compels the ResNet-18 to train an overall demonstration of the indoor scenario, where the regression head offers local coordinate-level accuracy and the classification head offers global "room level" context.

## 4.8 Chapter summary

The chapter provides comprehensive information on the proposed indoor positioning framework, including AP deployment, environment modeling, feature extraction and fusion, ray-tracing-based data generation, and the training and design of a dual-head ResNet-18 model. The training method, loss formulation, and estimation parameters were described to promote both regression-based coordinate positioning and zone-level classification, as illustrated in fig 16. The efficiency of the proposed framework is evaluated in the following chapter through a complete validation by applying empirical localization and classification performance parameters.

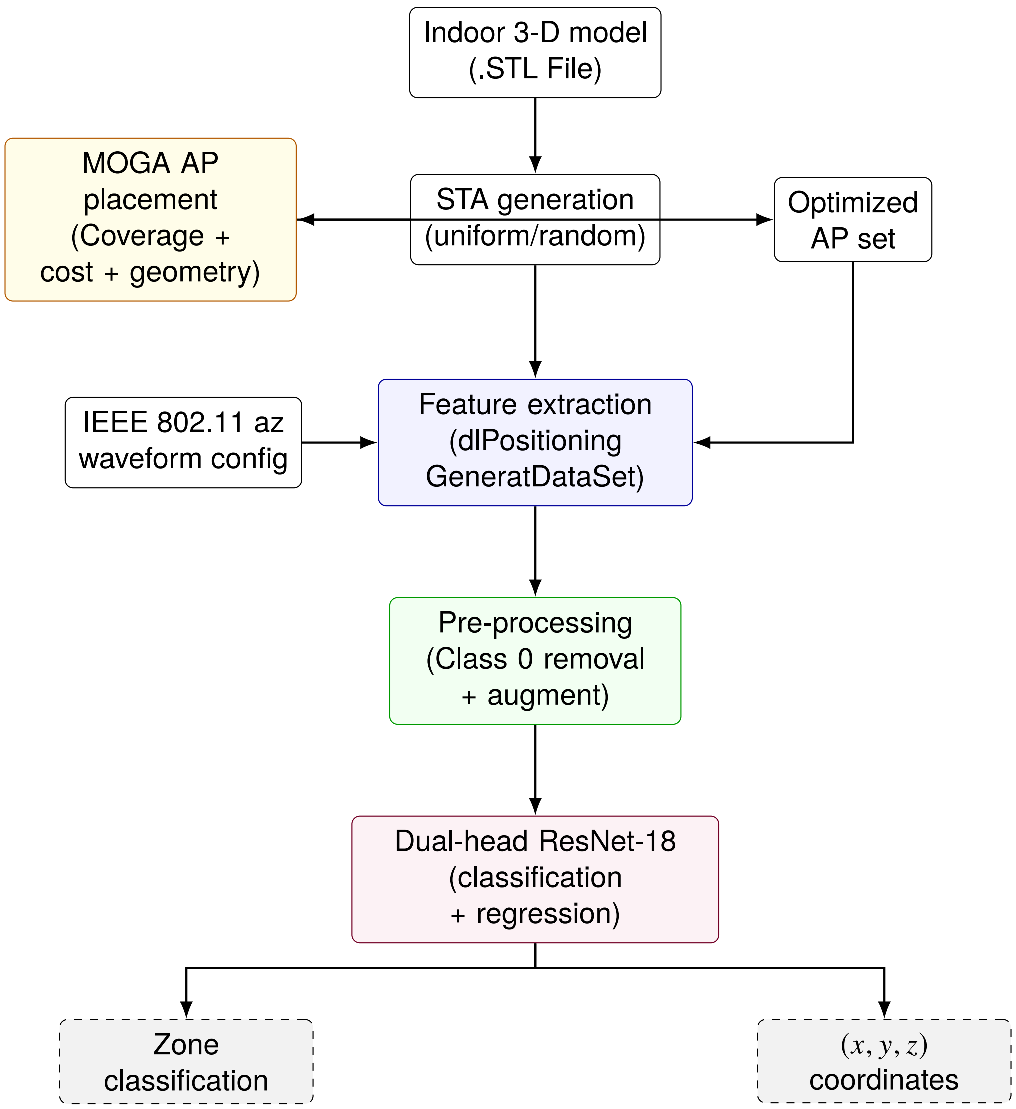


Figure 16: Comprehensive end-to-end framework pipeline. The system integrates BIM modeling, MOGA optimization, and multi-task learning for precise localization.

# 5 Validation

The chapter estimates the performance of the proposed indoor positioning framework established in the earlier chapters. The trained dual-head ResNet-18 framework is evaluated using both regression and classification metrics on unexplored validation data. Positioning accuracy is measured leveraging RMSE and CDF; however, classification performance is assessed utilizing accuracy and confusion matrices.

## 5.1 Validation methodology

Table 5: Dataset configuration parameters for indoor positioning

| ID | STA Sep. (m) | BW (MHz) | Antenna | Material | Dist. | STL Design (m) |
|---|---|---|---|---|---|---|
| 1 | 0.7 | CBW160 | 4×1 | Plasterboard | Random | 39 × 15 |
| 2 | 0.7 | CBW160 | 2×1 | Plasterboard | Random | 39 × 15 |
| 3 | 0.7 | CBW160 | 1×1 | Plasterboard | Random | 39 × 15 |
| 4 | 0.50 | CBW40 | 4×1 | Concrete | Uniform | 39 × 15 |
| 5 | 0.70 | CBW40 | 4×1 | Concrete | Uniform | 39 × 15 |
| 6 | 0.5 | CBW80 | 4×1 | Metal | Uniform | 39 × 15 |
| 7 | 0.5 | CBW160 | 4×1 | Metal | Uniform | 39 × 15 |
| 8 | 0.5 | CBW40 | 4×1 | Metal | Uniform | 39 × 15 |

After introducing the system framework, signal modeling, multi-modal characteristic fusion strategy, and deep learning framework in the prior chapters, the next step is to validate the proposed IEEE 802.11 az-based hybrid indoor positioning system within a configurable, reproducible experimental setup. The validation step aims to methodologically assess the impact of bandwidth, STA spatial density, material components, antenna array design, and distribution strategies on both regression and classification performance.

The performance framework consists of three channel bandwidth designs, such as CBW40, CBW80, and CBW160, to assess the influence of temporal resolution on localization accuracy. Antenna array designs of 1 × 1, 2 × 1, and 4 × 1 are employed to analyze the impact of spatial diversity and angular resolution on geometric reliability. To examine the effects of fingerprint density and spatial overlap, STA separation distances of 0.5 m, and 0.7 m are presented. To capture actual multi-path behavior, three surface materials are integrated into the ray-tracing propagation representation: concrete, metal, and plasterboard. Uniform grid-based and random distribution are the two STA placement strategies considered, allowing both detailed fingerprinting and theoretical deployment to be explored. Moreover, two characteristic strategies are estimated, comprising of CIR-only baseline and a multi-modal fusion technique incorporating RSSI, ToF, AoA, and CIR, illustrated in Table 5.

Using a 3-D BIM environment, validation is performed. The primary environment corresponds to a larger actual indoor design measuring 39 m × 15 m. The larger layout enables scalability assessment in transmission scenarios with improved multi-path diversity and significant NLOS transmission.

Performance is evaluated for regression error parameters such as RMSE, for continuous coordinate estimation, and for discrete zone classification using a confusion matrix.

Table 5 outlines the dataset configurations utilized during validation, comprising STA separation, bandwidth, antenna setup, material type, distribution, and STL design dimensions.

### 5.1.1 Evaluation of path loss extremes and blocking of signal

Table 6: Representative path loss and RSSI data for an individual station point

| **AP id** | **Path loss (dB)** | **RSSI (dBm)** | **Propagation status** |
|---|---|---|---|
| 1 – 6 | 300.00 | -280.00 | Obstructed / Null |
| 7 | 71.69 | -51.69 | Valid Connection |
| 13 | 69.29 | -49.29 | Valid Connection |
| 21 | 55.73 | -35.73 | Strong Connection |
| 35 | 46.54 | -26.54 | Strong Connection |
| 37 – 57 | 300.00 | -280.00 | Obstructed / Null |

Validation of the ray tracing modeling indicates a distinct difference between obstructed paths and LoS paths. Table 6 introduces RSSI and path loss values for one formal STA with respect to 56 optimized AP positions( random distribution with STA separation 0.7 m, along with plasterboard material, and fusion of signal characteristics). A path loss of 300 dB and an RSSI of -280 dB is revealed by most of the APs which includes IDs 1-6 and 37-56. Such values indicate entire signal is blocked resulting from structural components in the STL model, namely storage partitions and concrete walls. Valid signal propagation modeling is offered by APs 7, 13, 21, 28, and 35. Valid links range from -72.3 dB to -22.7 dB for RSSI, revealing moderate to strong connection. The effectiveness of the discrete optimization and ray-tracing framework is ensured by the difference between valid and blocked signals. Obstructed paths are considered as null characteristics, while a subset of APs fulfill the $K >= 3$ trilateration condition. The difference between valid and blocked signals enables the classification head to learn architectural signal patterns. Room-level classification accuracy is enhanced by learning such patterns.

### 5.1.2 Connection rate validation and K-coverage

Estimation of the 56-AP optimized layout reveals a comprehensive connection link of roughly 8.47% as indicated in Table 7. The value reveals suitable infrastructure density

Table 7: Global connectivity statistics for the optimized 56-AP layout

| Metric | Value | Technical significance |
|---|---|---|
| Total Stations ($N$) | 497 | Comprehensive spatial sampling |
| Mean APs per Station | 4.8 | Optimal trilateration density |
| Minimum APs per Station | 3 | Satisfies $K \geq 3$ requirement |
| Global Connection Rate | 8.47% | Minimizes co-channel interference |

for high positioning accuracy even though the percentage appears to be low. Every STA position preserve reliable connections with an average of 4 to 6 APs, fulfilling the $K >= 3$ constraint for trilateration. Signal saturation and co-channel interference is reduced by a controlled connection link, leading to a bigger SNR during CIR feature extraction.

## 5.2 Bandwidth impact assessment (group 3)

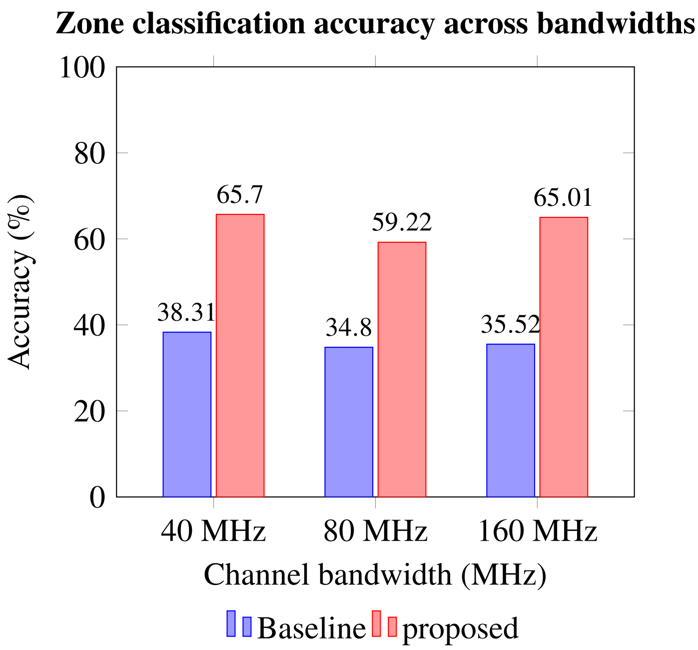


Figure 17: Classification performance comparison across IEEE 802.11 az bandwidths.

Wider channel bandwidth improves positioning system precision because higher frequencies provide finer temporal resolution for the CIR. The CBW160 configuration emerges as the superior choice because 160 MHz sampling allows the ResNet-18 model to distinguish the first signal path from complex indoor reflections with the greatest clarity.

To support comparable environmental geometry for CBW40, CBW80, and CBW160 configurations, the bandwidth impact is primarily assessed using the 39 m × 15 m controlled design. Stochastic theories demonstrate that higher bandwidth enhances time resolution of the CIR. In the delay domain, a broader bandwidth yields finer sampling, thereby improving ToF estimation and minimizing distance uncertainty. Enhanced delay resolution is predicted to result in smaller ranging error and sharper localization accuracy.

Validation results ensures the stochastic theories, as CBW160 configuration appears to be the best choice because 160 MHz sampling enables the ResNet-18 model to differentiate the direct path from dense indoor reflections with the maximum precision as in Figure 17. CBW80 configuration struggles to perform because of signal overlapping and enhanced susceptibility to multipath interference, whereas the high resolution reduces the positioning error to a 10.56 m RMSE. CBW40 requires the time-domain granularity important to solve precise coordinates as efficiently as CBW160 configuration. However CBW40 achieves high classification accuracy for room or zone classification. The proposed multimodal framework steadily surpasses the baseline across all bandwidth as the framework fuses the signal characteristics which preserve spatial reference points which raw signal characteristic lack.

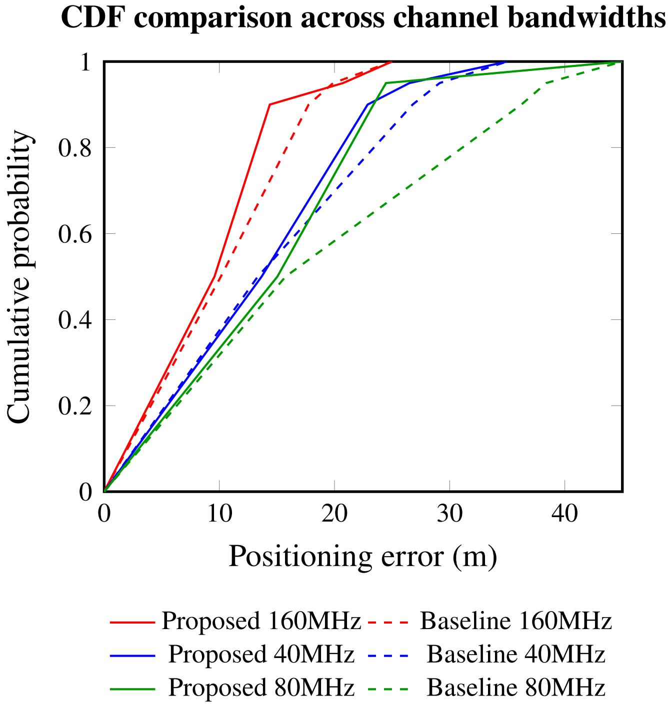


Figure 18: Cumulative distribution of positioning error representing the stable gains of multi-modal fusion across varying bandwidths.

The median error stays below 10 m and P90 error remains the lowest across all configurations as the best overall performance is provided by CBW160. CBW80 returns the unfavorable baseline results with a P90 error surpassing 36 m as per Figure 18, however the proposed multimodal fusion bounces back by minimizing the error to 23.38 m. CBW40 act as a intermediate configuration. Instead of enhancing the average position accuracy the proposed framework mostly enhances performance in worst case errors. Fusion of signal characteristics across all bandwidths confirms that the most of the test points fall in a fixed error interval compared to individual characteristic signal processing.

## 5.3 Effect of STA separation (group 2)

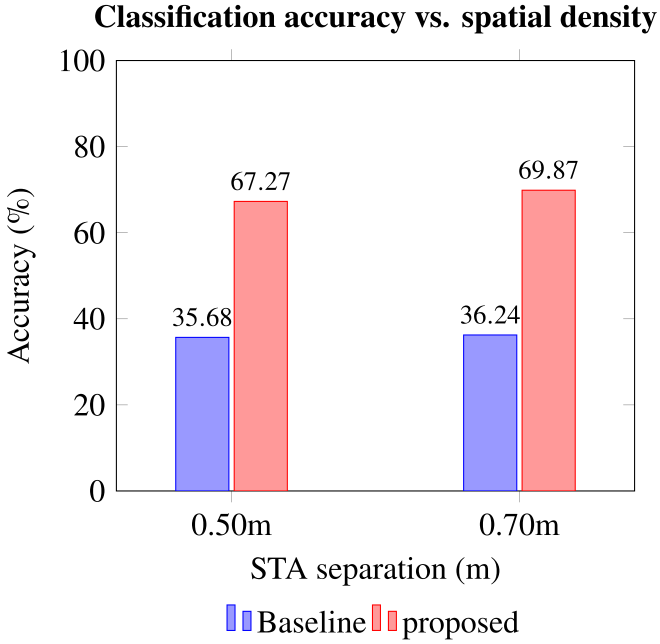


Figure 19: Zone classification performance across different STA separation.

STA separation straightly affect fingerprint density and the level of spatial detail in the dataset. Two separations are explored: 0.5 m demonstrating moderate density, and 0.7 m demonstrating lower density.

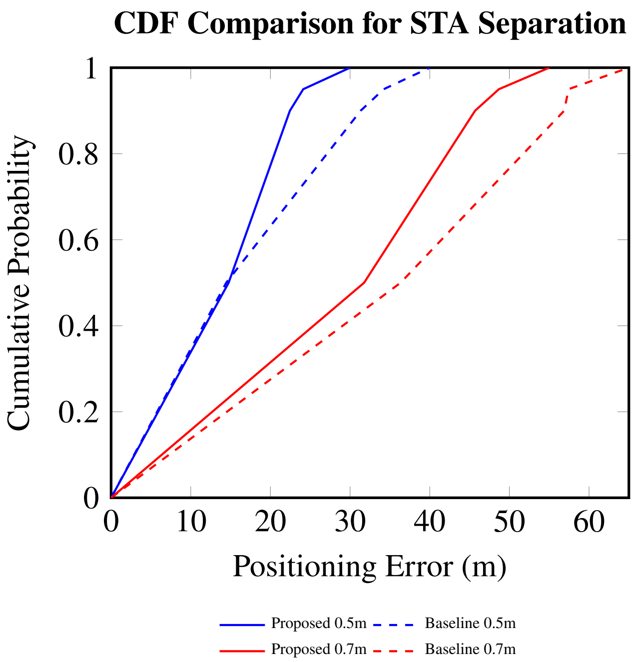


Figure 20: Cumulative error distribution validating 0.50 m as the superior density for coordinate precision.

The resolution of the fingerprinting database is directly determined by the spatial density of stations. A larger separation of 0.7 m preserve a high zone classification accuracy of

69.87% as indicated in the bar chart. However, drastically reduces coordinate precision. Proof still persists in the P90 and P95 regression errors, almost twice as much as to the 0.5 m configuration. By integrating signal characteristics information, performance degradation is reduced by the proposed framework. Geometric restrictions from such characteristics support balance for low sampling density. A STA separation of 0.5 m offers the best balance and obtains the coordinate precision necessary for autonomous navigation.

## 5.4 Impact of antenna array size (group 1)

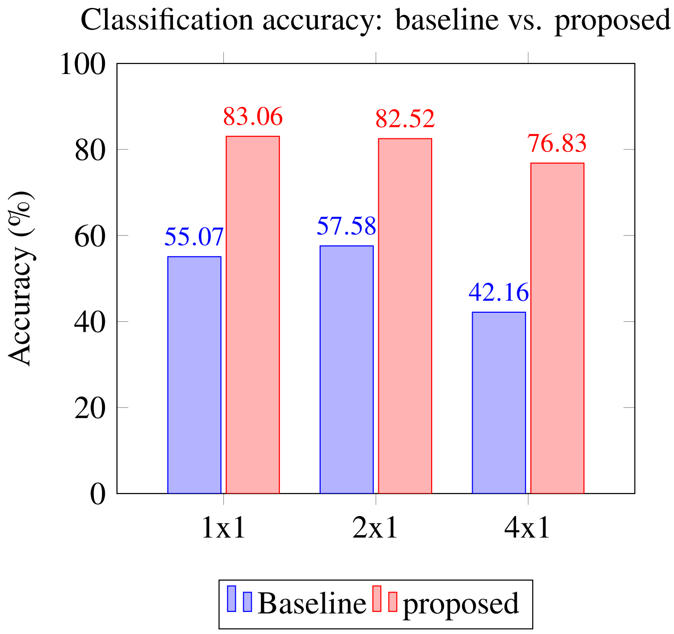


Figure 21: Classification accuracy comparison demonstrating reliable performance through multi-modal fusion.

The antenna array setup considerably impact angular diversity and geometric dilution across plasterboard and STA random distribution settings. Estimation over 1 x 1, 2 x 1, 4 x 1 configuration shows diverse performance trade-offs

The 1 x 1 configuration shows a single-input single-output (SISO) baseline. Performance remains reasonable under simple transmission scenarios, the need of spatial diversity causes larger multipath ambiguity, leading to a baseline RMSE of 9.19 m.

The 2x1 configuration enhances angular resolution with minimal hardware complexity. The 2 x 1 configuration offers a suitable balance between geometric reliability and computational load. 2 x 1 as the optimal configuration for the project ensured by the evaluation data, obtaining the lowest overall RMSE of 4.43 m and a high accuracy of 82.52%, see Figure 21. Without introducing high-dimensional feature noise into the tensor, 2 x 1 configuration delivers adequate spatial diversity for the MOGA to lower the GDOP.

The 4 x 1 configuration increases AoA estimation stability, the outcomes indicates a drop in both accuracy (76.83%) and regression accuracy (7.90 m RMSE) compared to 2 x

1 configuration. Performance gains are usually predicted in more complex multipath conditions; although, in the 4 x 1 configuration with a random distribution strategy, the enhanced dimensionality of the 4 x 1 CIR tensor looks like to add noise that impacts the dual-head reliability of the model.

## 5.5 Material sensitivity

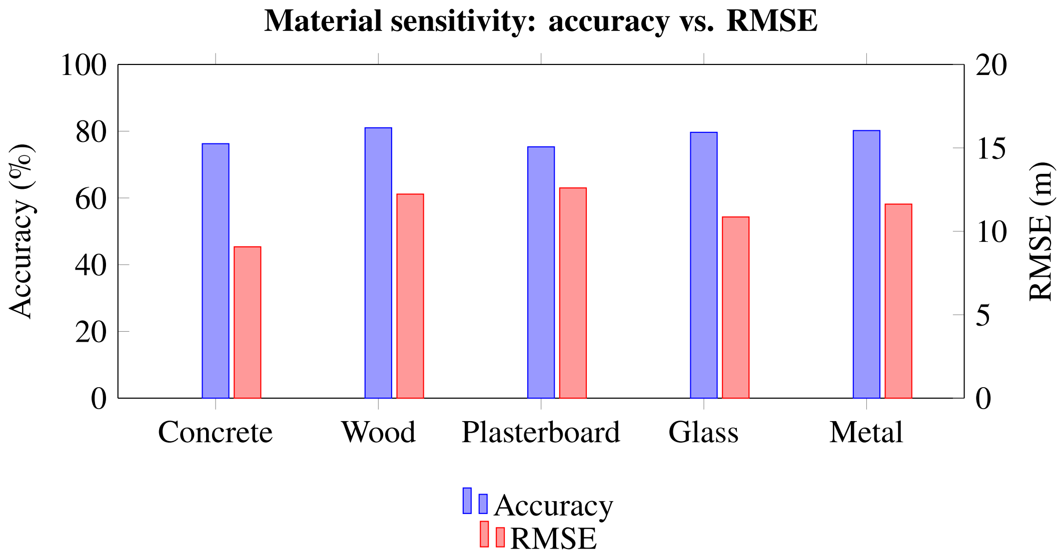


Figure 22: Performance trade-offs across common building materials using the 4×1 antenna configuration.

The most reliable material for indoor positioning appears to be concrete because concrete obtains the lowest RMSE of 9.07 m, representing robust signal reflections for coordinate regression when compared to metal (11.63 m) and glass (10.86 m). However metal and wood deliver a bit higher classification accuracy. On the other hand, plasterboard delivers the lowest performance with an RMSE of 12.60 m. Hence, concrete indicates the optimal material for preserving localization across the BIM layout in Figure 22.

## 5.6 Analysis of study groups

The final estimation is based on an integrated comparison of antenna setup, STA separation, and bandwidth, using both the baseline, which consists of an CIR-only characteristic, and the proposed multimodal fusion characteristic technique.

Table 8 presents a comparative assessment of antenna setup, STA separation, and bandwidth impact when applying both the baseline and proposed characteristic extraction strategies. The comparison is structured into three study groups to isolate the impact of individual design parameters while handling manageable experimental scenarios.

Table 8: Consolidated validation results: evaluation of antenna scaling, STA separation, and bandwidth using baseline (CIR-only) and proposed (CIR-ToF-AoA-RSSI) technique

| Study group | Variable | ID | Accuracy (%) | | RMSE (m) | |
|---|---|---|---|---|---|---|
| | | | Baseline | Proposed | Baseline | Proposed |
| **1. Antenna scaling** | 1×1 | 3 | 55.07 | 83.06 | 9.19 | 7.95 |
| | 2×1 | 2 | 57.58 | 82.52 | 7.33 | 4.43 |
| | 4×1 | 1 | 42.16 | 76.83 | 10.21 | 7.90 |
| **2. STA separation** | 0.50 m | 4 | 35.68 | 67.27 | 17.83 | 15.89 |
| | 0.70 m | 5 | 36.24 | 69.87 | 39.895 | 33.93 |
| **3. Bandwidth** | CBW40 | 6 | 38.31 | 65.70 | 15.45 | 14.87 |
| | CBW80 | 7 | 34.80 | 59.22 | 19.39 | 16.12 |
| | CBW160 | 8 | 35.52 | 65.01 | 11.22 | 10.56 |

The primary study group examines the influence of antenna-array setup. Within the study group, antenna setup varies across three configurations: 1 x 1, 2 x 1, and 4 x 1. The remaining parameters remain constant to facilitate comparative evaluation. STA separation is fixed at 0.7 m, channel bandwidth is stable at CBW160, material type is plasterboard, STA distribution strategy follows a random spatial placement, and the STL environment represents to the 39 m * 15 m layout. In the dataset configuration, uncertainty in performance is calculated using RMSE, while classification accuracy is computed using a confusion matrix. Results in Table 6 indicate that a structural variety and angular resolution enhance localization performance when temporal resolution is fixed.

The secondary study group explores the influence of STA separation. The group explores the influence of STA separation. In the present case, the antenna array setup is fixed at 4 × 1 to support high angular reliability. Channel bandwidth is constant at CBW40, material type is concrete, STA distribution strategy is uniform grid-based placement, and the STL environment represents the 39 m * 15 m large-scale layout. STA separation changes across two values: 0.5 m, and 0.7 m. Under such managed scenarios, the monitored differences in regression and classification performance increase fundamentally from adaptation in spatial sampling resolution and fingerprint density. Smaller separation increases spatial granularity and dataset density, whereas larger separation enhances zone-level diversity while limiting computational complexity. Table 6 describes how bandwidth impacts regression error and classification accuracy in a highly reflective environment.

The third study group explores the impact of bandwidth on difficult reflective scenarios. Antenna array setup is constant to confirm reliable angular diversity. STA separation remains fixed at 0.5 m, material type is metal, which corresponds to extreme multi-path, STA distribution strategy follows uniform placement, and the STL environment represents the 39 m * 15 m layout. Channel bandwidth ranges in between CBW40, CBW80 and

CBW160. In such scenarios, performance variations may be due to temporal resolution differences in the CIR and ToF estimations. Table 6 outlines how bandwidth scaling impacts regression error and classification accuracy in a highly reflective scenario.

Across all the study groups, dual characteristics extraction strategies are compared. The baseline configuration corresponds to the CIR-only technique, where the CIR magnitude serves as the sole characteristic for the dual-head ResNet-18 framework. However, the proposed configuration incorporates multi-modal fusion integrating RSSI, ToF, AoA, and CIR characteristics. The proposed technique consequently fuses power-domain, temporal, and angular data to enhance spatial fingerprint richness. Both regression and classification outcomes are produced by the equivalent neural network, confirming structural reliability across comparisons. Variation in performance between the baseline and the proposed framework, consequently, reflects the assistance of a multi-modal fusion characteristic rather than changes in the network architecture.

Table **??** concludes accuracy and RMSE values for both regression and classification throughout the study groups. Each row represents a particular managed experimental setup, and each column corresponds to the numerical assessment of the baseline and proposed framework. Performance adaptation across antenna-array setups reveals the influence of spatial diversity. Performance adaptation across bandwidth and temporal-resolution measures under reflective-material scenarios. Such systematic grouping confirms that the conclusions derived from Table 6 are significantly related to controlled parameter differences rather than to unnecessary environmental variations.

The structure of Table 6 consequently provides standardized validation of how spatial diversity, temporal resolution, fingerprint density, and characteristic fusion affect positioning performance in large-scale actual indoor scenarios.

## 5.7 Limitations

When compared to baseline model the proposed framework demonstrates strong localization robustness; yet, reliable centimeter-level accuracy continues to be a challenge in cluttered indoor environments. Limited signal bandwidth and complex multipath reflections available in the BIM environment is captured by 2 x 1 configurations with RMSE of 4.43 m. Propagation conditions produces a noise floor in the CIR, which decreases the positioning accuracy for minimal spatial differences. The discrete optimization strategy chooses optimized AP deployments, however limited grid resolution restrains achieving geometric precision. The proposed multimodal fusion enhances classification accuracy, while positioning accuracy relies mainly on grid spacing and signal resolution in the optimization process.

# 6 Conclusion

In this project, an infrastructure-supported localization framework leveraging the 802.11 az standard is designed to enable accurate, robust indoor positioning for autonomous mobile robots. The research introduced in this project represents that obtaining stable, sub-meter positioning in complex building environments needs a withdrawal from elementary, individual characteristic signal processing. Combining discrete optimization strategy for infrastructure deployment with multimodal deep learning fusion, the proposed framework address the basic challenges of signal deterioration and geometric dilution. By evaluating all the study groups the best configuration is 2 x 1 antenna scaling, 0.50 m STA separation, and CBW160 MHz bandwidth obtained a classification accuracy of 82.52% and a median positioning error of 4.43 m. 2 x 1 configuration enables the ResNet-18 model by applying fused RSSI, AoA, ToF and CIR information with highest resolutions, transforming raw signal information into stable indoor position information. To reduce positioning errors, strategic deployment of APs are optimized through MOGA and offers the important spatial diversity. The results suggest that IEEE 802.11 az has the potential to achieve centimeter-level accuracy when assisted by optimized infrastructure design and fusion-based techniques.

### 6.0.1 Future work

Future work will include validating the framework, using commercial 802.11 az hardware to evaluate the performance of 802.11 az-standard-based communication hardware in real-world scenarios. The optimization process can be enhanced by considering real-world factors, namely maintenance needs, operational hurdles, and equipment costs. The neural network can be improved by employing models that incorporate phase data, a transformer-based framework, and approaches that account for development to enable seamless positioning monitoring. Further enhancements may include broadening the system to measure comprehensive 3-D positions and orientations, and integrating wireless data with inertial and motion-sensor data (e.g., IMU odometry), thereby reducing drift and enhancing temporal reliability during robotic movements. In addition, transfer learning or meta-learning approaches may be combined so that the trained models can adapt to new floors or buildings with less data collection, thereby considerably reducing deployment effort.

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